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AIOB, MATH 5 Notes.

NUMERICAL METHODS FOR SCIENCE AND ENGINEERING

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#INTRODUCTION
Ch 01

Error

⇒ Numerical calculation can be in error due to the use of approximate values in the calculation.

The following definitions are used to measuring the errors.

1. Absolute error.

2. Relative error.

3. Percentage error.

4. Round-off error.

5. Truncation error.

Absolute error: (Aer)

Let,

x = Exact Value

x^* = Numerical Value.

$$\therefore \text{Aer} = |x - x^*|$$

Relative Error: (Per)

$$\text{Per} = \frac{|x - x^*|}{|x|}$$

Percentage error: (Per)

$$\text{Per} = \frac{|x - x^*|}{|x|} \times 100\%$$

Round-off error: (ROer)

$$\text{ROer} = \frac{2}{3} = 0.66666666\ldots \\ \approx 0.667$$

Truncation Error: (Ter)

Q:-

Then:-

$$e = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

$$e^{x^*} \approx 1 + x + \frac{x^2}{2!}$$

$$\therefore \text{Ter} = |e - e^{x^*}| = \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$$

$$|e^{10} - 10| = 10.9$$

(Ans) = small value

$$\frac{|e^{10} - 10|}{|10|} = 1.09$$

Basics In MATLAB

Matrices:-

⇒ All variable in MATLAB are treated as Matrices or arrays.

* A row vector entered as:-

⇒ $x = [1, 2, 3]$ or

⇒ $x = [1 \ 2 \ 3]$

* A Column Vector entered as:-

⇒ $y = [4; 5; 6]$ or

⇒ $y = [4 \ 5 \ 6]^T$

Note:- Semicolons (;) are used to separate the rows of a matrix.

⇒ Example of a 3 by 4 matrix is:-

$$B = [1, 2, 3, 4; 5, 6, 7, 8; 9, 10, 11, 12]$$

Programming in MATLAB:-

Loops:-

⇒ To repeat similar calculation several times we can use "for and while" loops.

Syntax of "for loops":-

```
for i = 1:K  
    commands...  
end
```

Syntax of "while loops":-

```
while statement == true  
    commands...  
end
```

Conditional:-

⇒ Conditional execution can be indicated by "if"

Statement

Syntax:-

```
if expression  
    commands...  
end
```

3) If there are more than 1 alternatives:-

Syntax:-

if expression₁
Commands...

else if expression₂
Commands...

elseif...

...
end

Example 1:-

⇒ Write a MATLAB codes for Calculating $n!$ using for loops.

>> clear

>> n = input('Enter a Number =');

fact = 1;

for i = 1:n

fact = fact * i;

end

>> fact

Solution of Linear System:

Good

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Solution of Linear System:

Solution Methods:-

- ① Gaussian Method using Partial pivoting.
- ② Gauss - Seidel Method.
- ③ Jacobie - Method.

☑ Difference between Total pivoting and partial pivoting:-

* Given that, $7x + 2y + 12z = 0$ $8x + 4y - 11z = 14$ $-2x + 5y + z = 10$	
Total pivoting	Partial pivoting
$12z + 7x + 2y = 0$ $-11z + 8x + 4y = 14$ $z - 2x + 5y = 10$	$8x + 4y - 11z = 14$ $7x + 2y + 12z = 0$ $-2x + 5y + z = 10$

Exercise-2# Gaussian Elimination with Partial Pivoting:-Example: 2.3:-

$$\Rightarrow 8I_1 + 3I_2 - 5I_3 = 3$$

$$10I_1 + 7I_2 + 2I_3 = 4$$

$$6I_1 + 4I_2 + 7I_3 = 8$$

(a) Solve the above system to find the currents in this circuit using Gaussian elimination with partial pivoting.

(b) Verify your result by direct substitution in the original equation.

(c) Write MATLAB code to solve by left division (backslash) operator.

$\Rightarrow$ In given:-

$$8I_1 + 3I_2 - 5I_3 = 3$$

$$10I_1 + 7I_2 + 2I_3 = 4$$

$$6I_1 + 4I_2 + 7I_3 = 8$$

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Now Using Partial pivoting (PP)

$$10I_1 + 7I_2 + 2I_3 = 4 \quad \text{--- (E1)}$$

$$8I_1 + 3I_2 - 5I_3 = 3 \quad \text{--- (E2)}$$

$$6I_1 + 4I_2 + 7I_3 = 8 \quad \text{--- (E3)}$$

operations	I_1	I_2	I_3	Constant	Equation	Pivot Equation
	10	7	2	4	E1	
	8	3	-5	3	E2	
	6	4	7	8	E3	
E1/10	1	$\frac{7}{10}$	$\frac{2}{10}$	$\frac{4}{10}$	E4	
E2/8	1	$\frac{3}{8}$	$-\frac{5}{8}$	$\frac{3}{8}$	E5	1st pivot
E3/6	1	$\frac{4}{6}$	$\frac{7}{6}$	$\frac{8}{6}$	E6	
E5 - E4	0	-0.325	-0.825	-0.025	E7	
E6 - E4	0	-0.033	0.967	0.033	E8	
E7/(-0.325)	0	1	2.528	0.076	E9	
E8/(-0.033)	0	1	-29.303	-28.27	E10	2nd pivot
E9 - E10	0	0	31.841	28.346	E11	
E11/31.841	0	0	1	1.11	E12	3rd pivot

Thus- $I_1 + \frac{7}{10} I_2 + \frac{2}{10} I_3 = -\frac{4}{10}$

$$0 + I_2 + 2.538 I_3 = 0.076$$

$$0 + 0 + I_3 = 1.11$$

So, $I_2 = 0.076 - 2.538 I_3$
 $= 0.076 - 2.538 \times 1.11$
 $= -2.741$

And- $I_1 = -\frac{4}{10} - \frac{7}{10} I_2 - \frac{2}{10} I_3$
 $= -\frac{4}{10} - \frac{7}{10} (-2.741) - \frac{2}{10} (1.11)$
 $= 2.0967$

$$\therefore (I_1, I_2, I_3) = (2.0967, -2.741, 1.11)$$

Justification:-

Put I_1, I_2, I_3 Values in given equation:-

$$\Rightarrow 8I_1 + 3I_2 - I_3 = 3$$

$$\Rightarrow 8 \times (2.0967) + 3 \times (-2.741) - 1 \times (1.11) = 3.0806 \approx 3$$

$$\Rightarrow 10I_1 + 7I_2 + 2I_3 = 4$$

$$\Rightarrow 10 \times (2.0967) + 7 \times (-2.741) + 2 \times (1.11) = 4 = 4$$

$$\Rightarrow 6I_1 + 4I_2 + 7I_3 = 8$$

$$\Rightarrow 6 \times (2.0967) + 4 \times (-2.741) + 7 \times 1.11 = 7.38 \approx 8$$

(C)

MATLAB code:-

```
>> clear;
>> A = [8, 3, -5; 10, 7, 2; 6, 4, 7]
>> B = [3; 4; 8]
>> Solution = A \ B
```

The Concept of backslash:-

⇒ We know,

Matrix form $\Rightarrow AX = B$

$$\Rightarrow A^{-1}AX = A^{-1}B$$

$$\Rightarrow X = A^{-1}B$$

In MATLAB Back-slash (\) represents the Inverse.

Exercise-25E-3(a)
mon

$$5x - 4y + 3z = 21$$

$$2x + 3y + 2z = 20$$

$$8x + 2y + z = 13$$

- (i) Solve the above linear system using Gaussian elimination with pivoting (partial/total)
- (ii) Justify your result direct substitution in original equations.
- (iii) Write MATLAB codes to solve by $\backslash$ division (backslash) operator.

 $\Rightarrow$ In given:-

$$5x - 4y + 3z = 21$$

$$2x + 3y + 2z = 20$$

$$8x + 2y + z = 13$$

(a)

using D.P:-
mon

$$8x + 2y + z = 13 \quad \text{--- (E1)}$$

$$5x - 4y + 3z = 21 \quad \text{--- (E2)}$$

$$2x + 3y + 2z = 20 \quad \text{--- (E3)}$$

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Operation	x	y	z	Constant	Equation	Pivot Equation
	8	2	1	13	E1	
	5	-4	3	21	E2	
	2	3	2	20	E3	
E1/8	1	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{13}{8}$	E4	1st Pivot
E2/5	1	$-\frac{4}{5}$	$\frac{3}{5}$	$\frac{21}{5}$	E5	
E3/2	1	$\frac{3}{2}$	$\frac{2}{2}$	10	E6	
E5-E4	0	-1.05	0.475	2.57	E7	
E6-E4	0	1.25	0.875	8.375	E8	
E8	0	1.25	0.875	8.375	E8	
E7	0	-1.05	0.475	2.57	E7	
E8/1.25	0	1	0.7	6.7	E(10-1)	2nd pivot
E7/(-1.05)	0	1	-0.45	-2.45	E(11-1)	
E9 - E10	0	0	1.15	9.15	E11	
E11/1.15	0	0	1	7.95	E12	3rd Pivot

Thus:-

$$x + \frac{2}{8}y + \frac{1}{8}z = \frac{13}{8}$$

$$0 + y + 0.7z = 6.7$$

$$0 + 0 + z = 7.95$$

Therefore:-

$$y = 6.7 - 0.7z$$

$$= 6.7 - 0.7 \times 7.95$$

$$= 1.14$$

$$x = \frac{13}{8} - \frac{2}{8}y - \frac{1}{8}z$$

$$= \frac{13}{8} - \frac{2}{8}(1.14) - \frac{1}{8}(7.95)$$

$$= 0.3475$$

Hence,

$$x = 0.3475, y = 1.135, z = 7.95$$

(6)

Justification:-

put the value of x, y, z in given equation:-

$$5x - 4y + 3z = 21$$

$$\Rightarrow 5(0.3475) - 4(1.135) + 3(7.95) = 21.05 \approx 21$$

$$2x + 3y + 2z = 20$$

$$\Rightarrow 2(0.3475) - 3(1.135) + 2(7.95) = 20.001 \approx 20$$

$$8x + 2y + z = 13$$

$$\Rightarrow 8(0.3475) + 2(1.135) + 7.95 = 13.004 \approx 13$$

③

MATLAB Code:-

>> clear;

>> A = [5, -4, 3; 2, 3, 2; 8, 2, 1]

>> B = [21; 20; 13]

>> Solution = A \ B

Jacobi Iterative Method &
Gauss-Seidel Method

⇒ Consider a System of equations as:-

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Where $a_{11}, a_{12}, \dots, a_{32}, a_{33}$ are Co-efficients of x, y, z also a_{11}, a_{22}, a_{33} are diagonal elements.

Condition for diagonal dominant:-

$$|a_{11}| > |a_{12}| + |a_{13}|$$

$$|a_{22}| > |a_{21}| + |a_{23}|$$

$$|a_{33}| > |a_{31}| + |a_{32}|$$

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① If the above system is diagonally dominant then we can use Gauss-Seidel or Jacobi Method.

② Gauss-Seidel or Jacobi Method is known as iterative method.

③ If the values (initial) of x_0, y_0, z_0 are not given then consider $x_0 = y_0 = z_0 = 0$.

Jacobi iterative formulae

We know:-

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{11}x = b_1 - a_{12}y - a_{13}z$$

$$\therefore x = \frac{1}{a_{11}} [b_1 - a_{12}y - a_{13}z]$$

For n no.

$$x_{n+1} = \frac{1}{a_{11}} [b_1 - a_{12}y_n - a_{13}z_n]$$

$$y_{n+1} = \frac{1}{a_{22}} [b_2 - a_{21}x_{n+1} - a_{23}z_n]$$

$$z_{n+1} = \frac{1}{a_{33}} [b_3 - a_{31}x_{n+1} - a_{32}y_{n+1}]$$

Where,

$$n = 0, 1, 2, 3, \dots$$

Gauss-Seidel iterative formulae

We know:-

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$x = \frac{1}{a_{11}} [b_1 - a_{12}y - a_{13}z]$$

For n no.

$$x_{n+1} = \frac{1}{a_{11}} [b_1 - a_{12}y_n - a_{13}z_n]$$

$$y_{n+1} = \frac{1}{a_{22}} [b_2 - a_{21}x_{n+1} - a_{23}z_n]$$

$$z_{n+1} = \frac{1}{a_{33}} [b_3 - a_{31}x_{n+1} - a_{32}y_{n+1}]$$

Where:-

$$n = 0, 1, 2, 3, \dots$$

Exercise: 2:-

Ex: (4) (a)

⇒ Consider the system of equations as

$$x + 8y + 2z = 10,$$

$$2x - 5y + 7z = 4,$$

$$3x - 8y - z = 1$$

- (i) Reduce the above system to diagonally dominant form.
- (ii) Write the corresponding Gauss-Seidel and Jacobi iterative formulae.
- (iii) Complete two iterations to estimate the roots to 3 decimal points (d.p) with the initial values $x_0 = 0.25$, $y_0 = 0.8$, $z_0 = 0.75$.
- (iv) Justify your answer/result.
- (v) Write MATLAB codes using Left-division (back-slash) operator.
- (vi) Write MATLAB codes to iterate the above formula four times.

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⇒ In given:-

$$x + 8y + 3z = 10 \quad \text{--- (E1)}$$

$$3x - 5y + 7z = 4 \quad \text{--- (E2)}$$

$$3x - y - z = 1 \quad \text{--- (E3)}$$

①

⇒ Re-arranging the equations as (E3), (E1), (E1+E2)

$$3x - y - z = 1 \quad ; \quad |3| > |-1| + |-1|$$

$$x + 8y + 3z = 10 \quad ; \quad |8| > |1| + |3|$$

$$4x + 3y + 10z = 14 \quad ; \quad |10| > |4| + |3|$$

This is the required diagonally dominant form.

②

Jacobi iterative formula

$$1. \quad x_{n+1} = [1 + y_n + z_n] \times \frac{1}{3}$$

$$2. \quad y_{n+1} = [10 - x_n - 3z_n] \times \frac{1}{8}$$

$$3. \quad z_{n+1} = [14 - 4x_n - 3y_n] \times \frac{1}{10}$$

Gauss-Seidel iterative formula

$$1. \quad x_{n+1} = [1 + y_n + z_n] \times \frac{1}{3}$$

$$2. \quad y_{n+1} = [10 - x_{n+1} - 3z_n] \times \frac{1}{8}$$

$$3. \quad z_{n+1} = [14 - 4x_{n+1} - 3y_{n+1}] \times \frac{1}{10}$$

(iii)⇒ In given:-The initial values are:-

$$x_0 = 0.85$$

$$y_0 = 0.8$$

$$z_0 = 0.75$$

Gauss-Seidel Iteration:-For $n=0$;

$$x_{0+1} = \frac{1}{3}[1 + y_0 + z_0]$$

$$\therefore x_1 = \frac{1}{3}[1 + 0.8 + 0.75] = 0.850$$

$$y_{0+1} = \frac{1}{8}[10 - x_{0+1} - 3z_0]$$

$$\therefore y_1 = \frac{1}{8}[10 - 0.850 - 3 \times 0.75] = 0.863$$

$$z_{0+1} = \frac{1}{10}[14 - 4x_1 - 3y_1]$$

$$\therefore z_1 = \frac{1}{10}[14 - 4 \times 0.850 - 3 \times 0.863] = 0.801$$

Thus,

$$x_1 = 0.850,$$

$$y_1 = 0.863, \quad z_1 = 0.801$$

Now:-For $n=1$;

$$x_{1+1} = \frac{1}{3}[1 + y_1 + z_1]$$

$$x_2 = \frac{1}{3}[1 + 0.863 + 0.801] = 0.888$$

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$$y_2 = \frac{1}{8} [10 - x_2 - 3z_1]$$

$$\therefore y_2 = \frac{1}{8} [10 - 0.888 - 3 \times 0.801] = 0.839$$

$$z_2 = \frac{1}{10} [14 - 4x_2 - 3y_2]$$

$$\therefore z_2 = \frac{1}{10} [14 - 4 \times 0.888 - 3 \times 0.839] = 0.703$$

Therefore:-

$$(x_2, y_2, z_2) = (0.888, 0.839, 0.703)$$

Jacobi iteration:-

For $n=0$:-

$$x_1 = \frac{1}{3} [1 + y_0 + z_0] = \frac{1}{3} [1 + 0.8 + 0.75] = 0.850$$

$$y_1 = \frac{1}{8} [10 - x_0 - 3z_0] = \frac{1}{8} [10 - 0.8 - 3 \times 0.75] = 0.867$$

$$z_1 = \frac{1}{10} [14 - 4x_0 - 3y_0] = \frac{1}{10} [14 - 4 \times 0.8 - 3 \times 0.8] = 0.820$$

Thus:-

$$\therefore (x_1, y_1, z_1) = (0.850, 0.867, 0.820)$$

For $n=1$:-

$$x_2 = \frac{1}{3} [1 + y_1 + z_1] = \frac{1}{3} [1 + 0.867 + 0.820] = 0.895$$

$$y_2 = \frac{1}{8} [10 - x_1 - 3z_1] = \frac{1}{8} [10 - 0.850 - 3 \times 0.820] = 0.836$$

$$z_2 = \frac{1}{10} [14 - 4x_1 - 3y_1] = \frac{1}{10} [14 - 4 \times 0.850 - 3 \times 0.820] = 0.814$$

Therefore:-

$$(u_2, v_2, z_2) = (0.895, 0.836, 0.814)$$

iv

Justification of Gauss-Seidel method:-

put x_1, x_2, z_2 values in given equations:-

$$x + 8y + 3z = 10$$

$$\Rightarrow (0.888) + 8 \times (0.830) + 3 \times (0.793) = 0.970 \approx 10$$

$$3x - 5y + 7z = 4$$

$$\Rightarrow 3 \times (0.888) + 7 \times (0.793) + 5 \times (0.837) = 4.03 \approx 4$$

$$3x - y - z = 1$$

$$\Rightarrow 3 \times (0.888) - (0.839) - (0.793) = 1.032 \approx 1$$

- Justification of Jacobi method -

Put x_1, x_2, x_3 in given equation:-

$$\Rightarrow 0.875 + 8 \times 0.836 + 3 \times 0.814 = 10.241 \approx 10$$

$$\Rightarrow 3 \times (0.895) + 7 \times (0.814) - 5 \times (0.836) = 4.068 \approx 4$$

$$\Rightarrow 3 \times (0.895) - 0.836 - 0.814 = 1.008 \approx 1$$

(proved)

(V)

MATLAB Code:-

>> clear;

>> A = [1, 8, 3; 3, -5, 7; 9, -1, -1]

>> B = [10; 4; 1]

>> Solution = A \ B

(VI)

MATLAB code for Gauss-Seidel iterative formula:-

>> clear;

>> x(1) = 0.85;

>> y(1) = 0.8;

>> z(1) = 0.75;

>> iter(1) = 0;

>> for n = 1:4

>> iter(n+1) = n;

>> x(n+1) = (1 + y(n) + z(n)) / 3;

>> y(n+1) = (10 - x(n+1) - 3 * z(n)) / 8;

>> z(n+1) = (14 - 4 * x(n+1) - 3 * y(n+1)) / 10;

>> end

>> Solution = [iter', x', y', z']

here:- $x(1) = x_0$ $x(2) = x_1$ $y(1) = y_0$ $z(1) = z_0$

iter(1) = 0 iteration

∴ iter(2) = 1

So, iter(n+1) = n

MATLAB Code for Jacobi Iterative Formula:-

```
>> clear;
```

```
>> x(1) = 0.85;
```

```
>> y(1) = 0.8;
```

```
>> z(1) = 0.75;
```

```
>> iter(1) = 0;
```

```
>> for n = 1:4
```

```
>> iter(n+1) = n;
```

```
>> x(n+1) = (1 + y(n) + z(n))/3;
```

```
>> y(n+1) = (10 - x(n) - 3 * z(n))/8;
```

```
>> z(n+1) = (14 - 4 * x(n) - 3 * y(n))/10;
```

```
>> end
```

```
>> Solution = [iter', x', y', z']
```

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Ex: 9:-

Potato Chips	pop	Chocolate	cost(\$)
2	1	5	21.00
3	2	10	20.88
8	3	4	19.12

- (i) Construct the system of linear equations from the above problem.
- (ii) Reduce the system obtained into diagonally form.
- (iii) Write the corresponding Gauss-Seidel iteration formula.
- (iv) Compute two iterations to estimate the roots to 2.d.p using following initial values
 $x_0 = 0.2,$
 $y_0 = 1.25,$
 $z_0 = 1.65$
- (v) Verify your result
- (vi) Write MATLAB codes to iterate the above formula four times.

(i)

Let,Unit price of potato chips = x x x x pop = y x x x chocolate = z $\therefore$ System of equations:-

$$2x + 2y + 5z = 21 \quad \text{--- (E1)}$$

$$3x + 2y + 10z = 20.88 \quad \text{--- (E2)}$$

$$8x + 3y + 4z = 13.17 \quad \text{--- (E3)}$$

(ii)

Re-arranging the equations as (E3), (E1), (E2)

$$8x + 3y + 4z = 13.17; |8| > |4| + |3|$$

$$2x + 2y + 5z = 21; |5| > |2| + |2|$$

$$3x + 2y + 10z = 20.88; |10| > |3| + |2|$$

This is the required diagonally dominant form

iii

The Gauss-Seidel equation:-

$$x_{n+1} = \frac{1}{8} [13.17 - 3y_n - 4z_n]$$

$$y_{n+1} = \frac{1}{9} [21 - 2x_{n+1} - 5z_n]$$

$$z_{n+1} = \frac{1}{10} [20.88 - 3x_{n+1} - 2y_{n+1}]$$

iv

is given:-

$$x_0 = 0.2$$

$$y_0 = 1.25$$

$$z_0 = 1.65$$

Gauss-Seidel Iteration:-

For $n=0$;

$$x_1 = \frac{1}{8} [13.17 - 3 \times y_0 - 4z_0]$$

$$= \frac{1}{8} [13.17 - 3 \times 1.25 - 4 \times 1.65]$$

$$= 0.35$$

$$y_1 = \frac{1}{9} [21 - 2x_1 - 5z_0]$$

$$= \frac{1}{9} [21 - 2 \times 0.35 - 5 \times 1.65]$$

$$= 1.03$$

$$z_1 = \frac{1}{10} [20.88 - 3x_1 - 2y_1]$$

$$= \frac{1}{10} [20.88 - 3 \times 0.35 - 2 \times 1.33]$$

$$= 1.71$$

$$\therefore (x_1, y_1, z_1) = (0.35, 1.33, 1.71)$$

For $n=2$

$$x_2 = \frac{1}{8} [13.17 - 3y_1 - 4z_1]$$

$$= \frac{1}{8} [13.17 - 3 \times 1.33 - 4 \times 1.71]$$

$$= 0.20$$

$$y_2 = \frac{1}{6} [21 - 2x_2 - 5z_1]$$

$$= \frac{1}{6} [21 - 2 \times 0.20 - 5 \times 1.71]$$

$$= 1.31$$

$$z_2 = \frac{1}{10} [20.88 - 3x_2 - 2y_2]$$

$$= \frac{1}{10} [20.88 - 3 \times 0.20 - 2 \times 1.31]$$

$$= 1.73$$

$$\therefore (x_2, y_2, z_2) = (0.20, 1.31, 1.73)$$

justification:-

$$2x + 9y + 5z = 21$$

$$\Rightarrow 2x(0.20) + 9x(1.31) + 5x(1.73) = 21.02 \approx 21$$

$$3x + 2y + 10z = 20.88$$

$$\Rightarrow 3x(0.20) + 2x(1.31) + 5x(1.73) \times 2 = 20.70 \approx 20.88$$

$$8x + 3y + 4z = 13.17$$

$$\Rightarrow 8(0.20) + 3x(1.31) + 4(1.73) = 13.17 = 13.17$$

(justified)

(VI)

MATLAB Code:-

```
>> clear;
```

```
>> x(1) = 0.2;
```

```
>> y(1) = 1.25;
```

```
>> z(1) = 1.65;
```

```
>> iter(1) = 0;
```

```
>> for n = 1:4
```

```
>> iter(n+1) = n;
```

```
>> x(n+1) = (13.17 - 3*y(n) - 4*z(n))/8;
```

```
>> y(n+1) = (21 - 2*x(n+1) - 5*z(n))/9;
```

$$\gg z(n+1) = (20.88 - 3 * x(n+1) - 2 * y(n+1)) / 10;$$

$\gg \text{end}$

$\gg \text{Solution} = [\text{iter}', n', y', z']$

VII

MATLAB Code:-

$\gg \text{clear};$

$\gg A = [2, 9, 5; 3, 2, 10, 8, 3, 4]$

$\gg B = [21.00; 20.88; 17.17]$

$\gg \text{Solution} = A \setminus B$

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Non-Linear Equations in One Variable

Chp 03

Non-Linear Equation in one network

$$x^3 + 4x - 3 = 0$$

$$x - e^{-x} = 0$$

$$\cos 2x - 2x + 1 = 0$$

} Non-Linear equation.

there are 5 (Five) Method:-

1. Number of real roots using graphical method
(only 1 method by which can determine number of roots)

2. Bisection Method:-

$$x_{n+1} = \frac{a_n + b_n}{2}$$

$$n = 1, 2, 3, \dots$$

3. Newton's Method/Newton-Raphson (N-R) method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$n = 0, 1, 2, 3, \dots$$

④ Secant Method

$$x_{n+2} = x_{n+1} - \frac{(x_{n+1} - x_n)}{f(x_{n+1}) - f(x_n)} \times f(x_{n+1})$$

where,

$n \geq 1$

⑤ Fixed-point Iteration Method:-

Number of real roots by
Graphical Method:-

~~~~~

 $\Rightarrow$  Let,

$f(x) = 0$  be a continuous function on  $(a, b)$

Consider,

$f(x) = 0$  as  $g_1(x) = g_2(x)$

Then, the total number of intersecting points on graph of  $g_1(x)$  and  $g_2(x)$   
= Number of real roots.

$$\therefore \text{Total root} = \text{No. of real root} + \text{No. of complex root}$$

Example 1:-

⇒ Consider  $f(x) = x^3 + 4x - 3 = 0$

- find the number of real roots of  $f(x) = 0$  by using graphical method.
- find the number of complex roots (if any)
- Use MATLAB function "roots" to find all the roots including complex roots

(a)

⇒ Given that:-

$$f(x) = 0$$

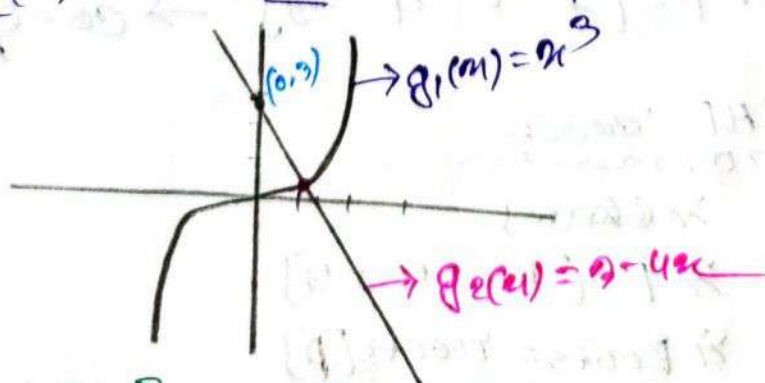
$$\Rightarrow x^3 + 4x - 3 = 0$$

$$\therefore x^3 = 3 - 4x$$

Let,

$$g_1(x) = x^3$$

$$g_2(x) = 3 - 4x$$



Since  $g_1(x)$  &  $g_2(x)$  intersect in 1 point  
So, No. of real roots = 1

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⇒ In given:-

$$x^3 - 4x - 3 = 0$$

∴ Total No. of roots = 3 [Since MAX power of  $x = 3$ ]

we know:-

No. of real roots = 1 [from 'a']

∴ Total roots = No. of real roots - No. of complex roots

$$\begin{aligned}\therefore \text{No. of complex roots} &= 3 - 1 \\ &= 2\end{aligned}$$

⇒ "roots" function works for only polynomial equation:

sol

$$f(x) = x^3 + 4x - 3$$

$$= x^3 + 0 \cdot x^2 + 4x - 3x^0 \rightarrow \text{Polynomial Equation.}$$

$$\therefore P = [1 \ 0 \ 4 \ -3] \rightarrow \text{co-efficient}$$

' MATLAB code

.' mmmm

» clear;

» P = [1 0 4 -3]

» Roots = roots(P)

## # The Conditions of Roots Finding

⇒ Let  $f(x) = 0$  be continuous on  $[a, b]$

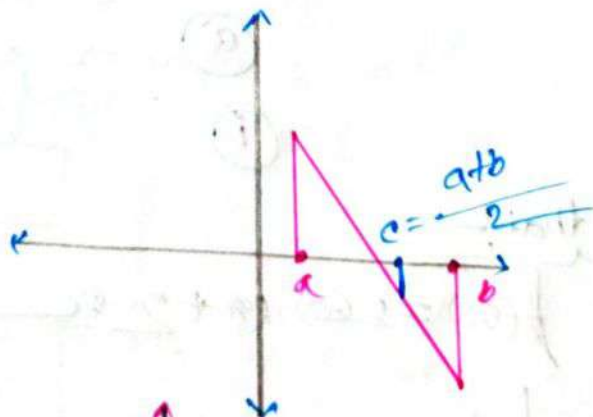
Then  $f(x) = 0$  has a root on  $(a, b)$  if

$$f(a) \cdot f(b) < 0$$

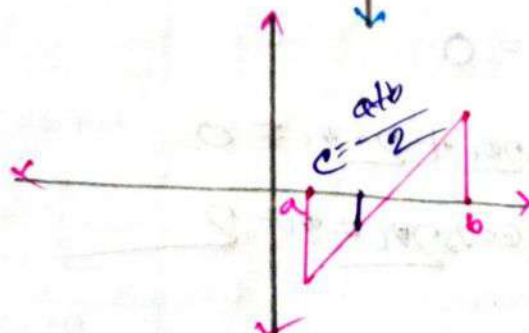
⇒ If  $f(a) \cdot f(b) \neq 0$  then we use  $f'(a) \cdot f'(b) < 0$

## # Bisection Method:-

(i) if  $a = +ve$   
 $b = -ve$



(ii) if  $a = -ve$   
 $b = +ve$



# Example : 1:-

Given that  $f(x) = 2\cos 2x + 2 - x$

- a) (i) find the number of roots of the equation  $f(x) = 0$
- (ii) For each root, find an interval where it lies.
- (b) (i) Show that the equation  $f(x) = 0$  has a root in  $[0.8, 1.6]$
- (ii) Use Bisection method three times to find the new smaller interval and root
- c) Write MATLAB code using "`fzero(fun,[a,b])`" or "`fzero(F,[a,b])`"

a

i

Given that:-

$$f(x) = 2\cos 2x + 2 - x$$

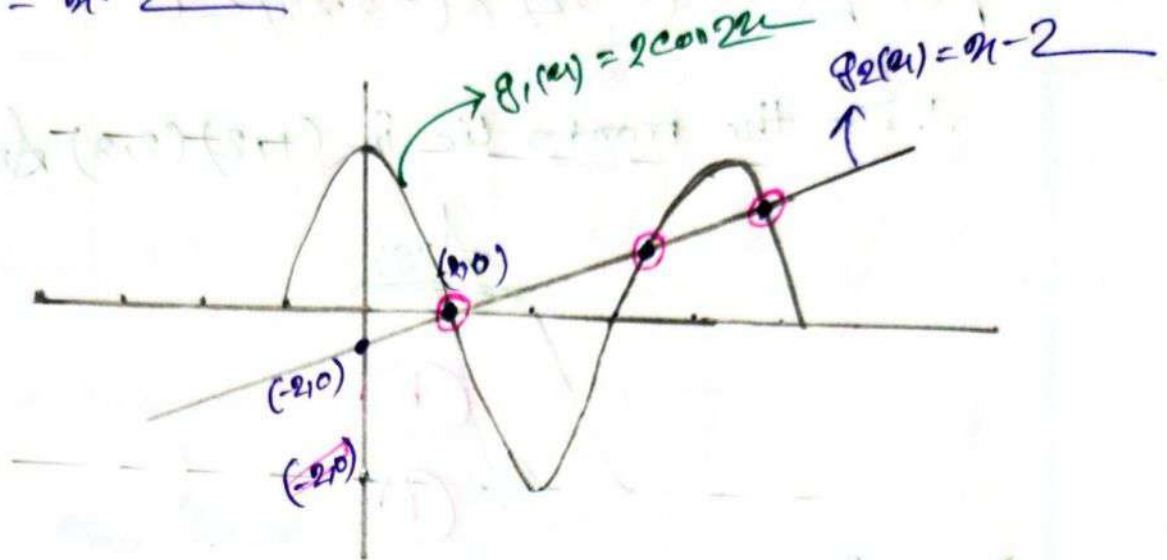
Here,  $f(x) = 0$

$$\therefore 2\cos 2x + 2 - x = 0$$

$$\text{so, } 2\cos 2x = x - 2$$

Let  $g_1(x) = 2 \cos 2x$

$g_2(x) = x - 2$



$\therefore$  no. of real roots = 3

Since  $g_1(x)$  &  $g_2(x)$  intersect in three points.

ii

$\Rightarrow$  we know:-  
more

The condition of roots finding  
more more

$f(a) \times f(b) < 0$   $\leftarrow$  ①

| $x$ | $f(x)$ |
|-----|--------|
| -2  | 2.69   |
| -1  | 2.17   |
| 0   | 4      |
| 1   | 0.167  |
| 2   | -1.30  |
| 3   | 0.12   |
| 4   | -2.29  |

Now,

put the  $(a, b)$  values in eq ①

1.  $f(-2) \times f(2) = 2.69 \times -1.30 \neq 0$

2.  $f(-1) \times f(0) = 2.17 \times 4 \neq 0$

3.  $f(0) \times f(1) = 4 \times 0.167 \neq 0$

4.  $f(1) \times f(2) = 0.167 \times -1.30 < 0$

$$\hookrightarrow f(2) \times f(3) = (-1.31) \times (0.92) < 0$$

$$\hookrightarrow f(3) \times f(4) = (0.92) \times (-2.29) < 0$$

$\therefore$  So the roots lie in  $(1, 2)$   $(2, 3)$  and  $(3, 4)$

Ans:-

(b)

(i)

$\Rightarrow$  In given  
now

The interval is  $[0.8, 1.6]$

Thus-

$$f(0.8) = 2 \cos 2x + 2 - x$$

$$= 2 \cos 2 \times 0.8 + 2 - 0.8$$

$$= 1.14$$

$$f(1.6) = 2 \cos 2 \times 1.6 + 2 - 1.6$$

$$= -1.59$$

we know-

$$f(a) \times f(b) < 0 \quad \text{--- (1)}$$

Then,  $f(x) = 0$  has a roots in  $(0.8, 1.6)$

Now put the value in eq. (1)  $\Rightarrow$

$$f(0.8) \times f(1.6) = (1.14) \times (-1.59) < 0$$

Thus,  $f(x) = 0$  has a root in  $[0.8, 1.6]$

Ans:-

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(ii)

| a   | b   | $c = \frac{a+b}{2}$ | $f(a)$ | $f(b)$ | $f(c)$ | Update  | New interval |
|-----|-----|---------------------|--------|--------|--------|---------|--------------|
| 0.8 | 1.6 | 1.2                 | 2.14   | -1.59  | -0.67  | $b = c$ | (0.8, 1.2)   |
| 0.8 | 1.2 | 1                   | 2.14   | -0.67  | 0.16   | $a = c$ | (1, 1.2)     |
| 1   | 1.2 | 1.1                 | 0.16   | -0.67  | -0.27  | $b = c$ | (1, 1.1)     |

Therefore New Smaller interval = (1, 1.1)

Ans. Roots/Root =  $\frac{a+b}{2}$

$$= \frac{1+1.1}{2}$$

$$= 1.05$$

(3)

MATLAB Code:

```
>> clear;
>> Fun = @(x) (2*cos(2*x) + 2 - x);
>> Solution = fzero(Fun, [0.8, 1.6])
```

The Student  
→ to define a  
function  
Fun = @(x) Equations

## # Newton's Method! -

### # Formula:-

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$n = 0, 1, 2, 3, \dots$$

### # Example: 1:-

⇒ Consider the function  $f(x) = \sin x + 3x - 1$

① perform two iteration using Newton-Raphson/Newton's method formula for finding root near  $x=0$  (or  $x_0=0$ )

② Write MATLAB code using "fzero" for  $[0, 1]$

⇒ ~~is~~ given:-

$$f(x) = \sin x + 3x - 1$$

$$\therefore f'(x) = \cos x + 3$$

Now,

At  $x=0$ ;

$$f(0) = \sin 0 + 3 \times 0 - 1 = -1$$

$$f'(0) = \cos 0 + 3 = 1 + 3 = 4$$

We know:-

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\therefore x_{n+1} = x_n - \frac{\sin(x_n) + 3x_n - 1}{\cos(x_n) + 3} \quad \text{--- (1)}$$

∴ 1st Iteration : 1:-

∴  $n=0$  :-

$$x_{0+1} = x_0 - \frac{\sin(x_0) + 3x_0 - 1}{\cos(x_0) + 3}$$

$$x_0 = 0$$

$$\begin{aligned} \therefore x_1 &= 0 - \frac{\sin 0 + 3 \times 0 - 1}{\cos 0 + 3} \\ &= 0 - \frac{-1}{4} \end{aligned}$$

$$\therefore x_1 = \frac{1}{4}$$

Now:- 2nd Iteration : 2:-

Now:-  $n=1$  :-

Substitute the value of  $n$  in eq (1) :-

$$\therefore x_{1+1} = x_1 - \frac{\sin x_1 + 3x_1 - 1}{\cos x_1 + 3}$$

$$\therefore x_2 = \frac{1}{4} - \frac{\sin(\frac{1}{4}) + 3 \times \frac{1}{4} - 1}{\cos(\frac{1}{4}) + 3}$$

$$\therefore x_1 = \frac{1}{4}$$

$$= 0.25065$$

(b)

MATLAB CODE:-

>> Clear;

>> Fun = @(x)(sin(x) + 3\*x - 1);

>> Solution = fzero(Fun, [0, 2])

Ans:-

## # Modified Newton-Raphson Method For Multiple Roots:-

### # Formula:-

$$x_{n+1} = x_n - m \times \frac{f(x_n)}{f'(x_n)}$$

$$n = 0, 1, 2, 3, \dots$$

$m$  = Number of roots.

### # Example: 1:-

⇒ The equation  $686x^3 - 735x^2 + 125 = 0$  has a double root near  $x = 0.5$  (or  $x_0 = 0.5$ ) Perform one iteration using modified Newton-Raphson method.

⇒ Is given:-

$$f(x) = 686x^3 - 735x^2 + 125$$

$$\begin{aligned} f'(x) &= 3 \times 686x^2 - 2 \times 735x \\ &= 2058x^2 - 1470x \end{aligned}$$

At Iteration 1:-

$$x_0 = 0.5$$

$$m = 2$$

We know:-

$$x_{n+1} = x_n - m \times \frac{f(x_n)}{f'(x_n)}$$

$$\therefore x_{n+1} = x_n - m \times \frac{686x_n^3 - 735x_n^2 + 125}{2058x_n^2 - 1470x_n} \quad \text{--- (1)}$$

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For  $n=0$ :-

$$x_1 = x_0 - m \times \frac{686x_0^3 - 705x_0^2 + 125}{2058x_0^2 - 1470x_0}$$

$$= 0.5 - 2 \times \frac{686(0.5)^3 - 705(0.5)^2 + 125}{2058(0.5)^2 - 1470(0.5)}$$

$$= 0.744$$

$$\therefore x_2 = 0.744$$

Ans:-

# Secant Method:-

# Formula:-

$$x_{n+2} = x_{n+1} - \frac{(x_{n+1} - x_n)}{f(x_{n+1}) - f(x_n)} \times f(x_{n+1})$$

Where,  $n \geq 1$

# Example 1:-

⇒ Consider  $f(x) = 2 \cos 2x + 2 - x = 0$  has a root in  $(1, 1.1)$   
or  $x_1 = 1, x_2 = 1.1$  Compute the value using Secant Method with accuracy tolerance  $\epsilon = 0.001$

⇒ here,

$$x_1 = 1$$

$$x_2 = 1.1$$

$$f(x) = 2 \cos 2x + 2 - x$$

⇒ In given:-

$$f(x) = 2 \cos 2x - x + 2$$

$$\therefore f(x_1) = f(1) = 2 \cos 2 \times 1 - 1 + 2 = 0.167$$

$$f(x_2) = f(1.1) = -0.277$$

we know:-

$$x_{n+2} = x_{n+1} - \frac{x_{n+1} - x_n}{f(x_{n+1}) - f(x_n)} \times f(x_{n+1})$$

For  $n=1$ :-

$$\begin{aligned} x_3 &= x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} \times f(x_2) \\ &= 1.1 - \frac{1.1 - 1}{-0.277 - 0.167} \times (-0.277) \end{aligned}$$

$$= 1.037$$

Also

$$\boxed{\varepsilon = |x_3 - x_2|}$$

$$= |1.037 - 1.1|$$

$$= 0.062$$

Now:- Again:-

$$f(x_3) = f(1.037) = 2 \cos(2 \times 1.037) - 1.037 + 2 = -0.00147$$

For  $n=2$ :-  
mon

$$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} \times f(x_3)$$

$$= 1.037 - \frac{1.037 - 1.1}{(-0.00147) - (-0.277)} \times (-0.00147)$$

$$= 1.036$$

also,

$$\epsilon = |x_4 - x_3|$$

$$= |1.036 - 1.037|$$

$$= 0.001$$

$$\therefore x_3 = 1.037$$

$$x_4 = 1.036$$

Ans:-

# Fixed point Iteration Method:-

$$\text{If } f(x) = 0$$

$$\therefore x = g(x)$$

$$\text{So, } x_{n+1} = g(x_n); \quad n = 0, 1, 2, \dots$$

For

Converge  $\rightarrow$  finite value.

Diverge  $\rightarrow$  Infinite value.

Formula:-

① if  $|g'(x_0)| < 1$ , then the root will Converge.  
to find the point ob

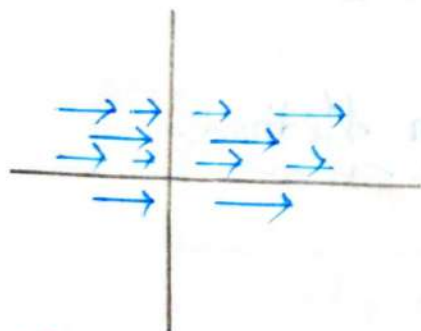
also:

(a) if  $-1 < g'(x_0) < 0$ , then Converge with oscillation.

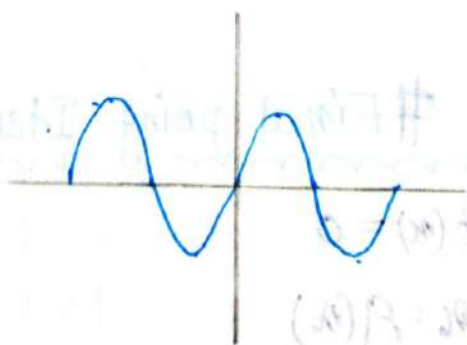
(b) if  $0 < g'(x_0) < 1$ , then Converge without oscillation.

② if  $|g'(x_0)| \geq 1$  then the root will Diverge.

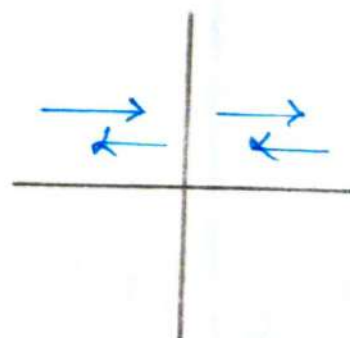
Ex:-



Converge without  
oscillation



Converge with  
Oscillation



Diverge.

### # Example 1:-

⇒ The equation  $x^3 + 2x - 5 = 0$  has a root near  $x = 1.4$  using the following iteration formula. Comment on the result and perform one iteration.

(a)  $x_{n+1} = \frac{1}{2}(5 - x_n^3)$

(b)  $x_{n+1} = (5 - 2x_n)^{\frac{1}{3}}$

(c)  $x_{n+1} = \frac{2x_n^3 + 5}{3x_n^2 + 2}$

⇒ In given:-

$x_{n+1} = \frac{1}{2}(5 - x_n^3)$  ————— (i)

Where,  $n = 0, 1, 2, \dots$

Let,

$q(x_n) = \frac{1}{2}(5 - x_n^3)$

∴  $q(x_0) = \frac{1}{2}(5 - x_0^3)$

So,  $q'(x_0) = \frac{1}{2} \times -3x^2 = -\frac{3}{2}x^2$  ————— (ii)

Now,

put  $x_0 = 1.4$  in eq (ii) ⇒

∴  $q'(1.4) = -\frac{3}{2}(1.4)^2$   
 $= -2.94$

So,

$|q'(x_0)| = |q'(1.4)| = |-2.94| = 2.94 > 1$

Then the root will diverge near  $x_0 = 1.4$

(6)

In given—

$$x_{n+1} = (5 - 2x_n)^{\frac{1}{3}} \quad \text{--- (1)}$$

Where,  $n = 0, 1, 2$ 

Let,  $g(x) = (5 - 2x)^{\frac{1}{3}}$

$$\therefore g'(x) = (5 - 2x)^{-\frac{2}{3}}$$

So,  $g'(x) = -\frac{2}{3}(5 - 2x)^{-\frac{2}{3}} \quad \text{--- (11)}$

Now, put  $x = 1.4$  in eq (11) =

$$\therefore g'(x) = -\frac{2}{3}(5 - 2 \times 1.4)^{-\frac{2}{3}}$$

$$= -0.304$$

So,  $|g'(1.4)| = |-0.304| = 0.304 < 1$

$\therefore$  The root will be Converge with Oscillation.  
as  $g'(1.4)$  is  $-1 < g'(1.4) < 0$

Now, Iteration : 01 :-

put  $n = 0$  in eq (1) =

$$x_1 = (5 - 2x_0)^{\frac{1}{3}}$$

$$= (5 - 2 \times 1.4)^{\frac{1}{3}}$$

$$= 1.30050$$

$\therefore$  The root in  $x_1 = 1.30050$  near  $x = 1.4$

②

Is given:-

$$x_{n+1} = \frac{2x_n^3 + 5}{3x_n^2 + 2} \quad \text{--- (1)}$$

Let,  $g(x) = \frac{2x^3 + 5}{3x^2 + 2}$

$$\therefore g(x) = \frac{2x^3 + 5}{3x^2 + 2}$$

$$\therefore g'(x) = \frac{(3x^2 + 2) \times 6x^2 - (2x^3 + 5) \times 6x}{(3x^2 + 2)^2} \quad \text{--- (ii)}$$

Put  $x = 1.4$  in eq (ii) =

$$\therefore g'(x) = \frac{\{3 \times (1.4)^2 + 2\} \times 6 \times (1.4)^2 - \{2 \times (1.4)^3 + 5\} \times 6 \times (1.4)}{\{3 \times (1.4)^2 + 2\}^2}$$

$$= 0.0735$$

$$\therefore |g'(x)| = |g'(1.4)| = 0.0735$$

$\therefore$  The root will Converge without oscillation.

Iteration 01:-

Put  $n = 0$  in eq (1) =

$$x_1 = \frac{2x_0^3 + 5}{3x_0^2 + 2} = \frac{2 \times (1.4)^3 + 5}{3 \times (1.4)^2 + 2} = 1.3300$$

$\therefore$  The root is  $x_1 = 1.3300$  near  $x = 1.4$

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# Non-Linear System of Equations!

Choi!

## # Non-Linear System of eq:-

1. Newton-Raphson Method.

2. Fixed point Iteration Method.

## # Newton-Raphson Method:-

Formula:-

$$1. X_{n+1} = X_n - [J(X_n)]^{-1} F(X_n)$$

Where,

$$X_{n+1} = \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix},$$

$$X_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix}$$

$$J(X) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix}$$

$$F(X_n) = \begin{pmatrix} f_1(x_n, y_n) \\ f_2(x_n, y_n) \end{pmatrix}$$

2. if  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  then

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

(11)

# Example : 1 :-

⇒ Consider the system of non-linear equation

$$x^2 + y = 1, \quad y^2 - 2x + 4 = 0$$

(a) Find the jacobian matrix  $J(x, y)$  for the above system.

(b) Evaluate the inverse of the jacobian matrix at  $(1.4, -1)$

(c) Write down the iteration formula using Newton Raphson method.

(d) perform one iteration to estimate root to start with  $(x_0, y_0) = (1.4, -1)$

(e) Use MATLAB function "fsolve" to find root.

⇒ On given :-

$$\begin{aligned} x^2 + y &= 1 \\ \therefore x^2 + y - 1 &= 0 \end{aligned} \quad \begin{aligned} y^2 - 2x + 4 &= 0 \\ \therefore y^2 - 2x + 4 &= 0 \end{aligned}$$

Let,  $f_1(x, y) = x^2 + y - 1$  ————— (i)

$f_2(x, y) = y^2 - 2x + 4$  ————— (ii)

(a)

we know:-

$$J(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}$$

$$= \begin{pmatrix} 2x & 1 \\ -2 & 3y^2 \end{pmatrix}$$

(b)

At (1.4, -1):-

$$J(1.4, -1) = \begin{pmatrix} 2 \times 1.4 & +1 \\ -2 & 3 \times (-1)^2 \end{pmatrix}$$

$$= \begin{pmatrix} 2.8 & 1 \\ -2 & 3 \end{pmatrix}$$

$$\therefore |J(1.4, -1)| = 8.4 + 2 = 10.4$$

So, the Inverse Matrix is:-

$$J^{-1}(1.4, -1) = \frac{1}{10.4} \begin{bmatrix} 3 & -1 \\ 2 & 2.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.28 & -0.096 \\ 0.19 & 0.269 \end{bmatrix}$$

②

We know:-

The iteration formula:-

$$X_{n+1} = X_n - [J(X_n)]^{-1} F(X_n)$$

$$\Rightarrow \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} x_n \\ y_n \end{pmatrix} - [J(X_n)]^{-1} \begin{pmatrix} f_1(x_n, y_n) \\ f_2(x_n, y_n) \end{pmatrix}$$

$$\therefore \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} x_n \\ y_n \end{pmatrix} - [J(X_n)]^{-1} \begin{pmatrix} x_n^2 + y_n - 1 \\ y_n^3 - 2x_n + 4 \end{pmatrix} \quad \text{--- (iii)}$$

③

Iteration 0:-

put  $n=0$  in (iii) eq. =

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} - [J(X_n)]^{-1} \begin{pmatrix} x_0^2 + y_0 - 1 \\ y_0^3 - 2x_0 + 4 \end{pmatrix} \quad \text{--- (iv)}$$

put  $x_0, y_0$  value in eq. (iv) =

$$\begin{aligned} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} 1.4 \\ -1 \end{pmatrix} - \begin{pmatrix} 0.28 & -0.09 \\ 0.19 & 0.26 \end{pmatrix} \times \begin{pmatrix} (1.4)^2 + (-1) - 1 \\ (-1)^3 - 2 \times 1.4 + 4 \end{pmatrix} \\ &= \begin{pmatrix} 1.4 \\ -1 \end{pmatrix} - \begin{pmatrix} 0.28 & -0.09 \\ 0.19 & 0.26 \end{pmatrix} \times \begin{pmatrix} -0.04 \\ 0.2 \end{pmatrix} \end{aligned}$$

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$$\begin{aligned} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} 1.4 \\ -1 \end{pmatrix} - \begin{pmatrix} -0.01 + 0.018 \\ -7.6 \times 10^{-3} + 0.052 \end{pmatrix} \\ &= \begin{pmatrix} 1.4 \\ -1 \end{pmatrix} - \begin{pmatrix} -0.028 \\ 0.044 \end{pmatrix} \\ &= \begin{pmatrix} 1.42 \\ -1.044 \end{pmatrix} \end{aligned}$$

∴ The roots are  $x_1 = 1.42$ ,  $y_1 = -1.044$

Ⓔ

MATLAB CODE:-

>> clear;

>> F = @(x) (x(1)^2 + x(2) - 1; x(2)^3 - 2\*x(1) + 4);

>> X0 = [1.43; -1];

>> Solution = solve(F, X0)

Note:-

$x = x(1)$

$y = x(2)$

## # Fixed-point Iteration Method:-

Let,

$$f_1(x, y) = 0,$$

$$f_2(x, y) = 0$$

Jacobi Method

$$\text{1. } x_{n+1} = g_1(x_n, y_n)$$

$$\text{2. } y_{n+1} = g_2(x_n, y_n)$$

Gauss-Seidel Method

$$\text{1. } x_{n+1} = g_1(x_n, y_n)$$

$$\text{2. } y_{n+1} = g_2(x_{n+1}, y_n)$$

Formula:-

$$\text{1. if } \left| \frac{\partial g_1(x_0, y_0)}{\partial x} \right| + \left| \frac{\partial g_1(x_0, y_0)}{\partial y} \right| < 1 \quad \&$$

$$\left| \frac{\partial g_2(x_0, y_0)}{\partial x} \right| + \left| \frac{\partial g_2(x_0, y_0)}{\partial y} \right| < 1 \quad \text{then}$$

the system will converge.

# Examples 1:-

⇒ Consider the System of equations  $y = x^2 - 5x + 3$ ,  $x^2 + 4y = 4$

(a) A fixed point iteration formula is suggested to estimate root at  $(x_0, y_0)$

$$x_{n+1} = \frac{1}{5}(x_n^2 - y_n + 3)$$

$$y_{n+1} = \frac{1}{10}(4 - x_n^2 - 4y_n^2 + 10y_n)$$

(i) Verify whether the above iterative formula will converge to the root near  $(x_0, y_0) = (0.4, 1.0)$ . if converge, then perform one iteration using Gauss-Seidel Method.

⇒ Is given:-

$$x_{n+1} = \frac{1}{5}(x_n^2 - y_n + 3) \quad \text{--- (i)}$$

$$y_{n+1} = \frac{1}{10}(4 - x_n^2 - 4y_n^2 + 10y_n) \quad \text{--- (ii)}$$

Let

$$x_{n+1} = g_1(x_n, y_n) = \frac{1}{5}(x_n^2 - y_n + 3) \quad \text{--- (iii)}$$

$$y_{n+1} = g_2(x_n, y_n) = \frac{1}{10}(4 - x_n^2 - 4y_n^2 + 10y_n) \quad \text{--- (iv)}$$

Now Diff (iii) & (iv) partially with respect to  $x$  &  $y$  we have:-

Let

$$g_1(x, y) = \frac{1}{5}(x^2 - y + 3)$$

$$g_2(x, y) = \frac{1}{10}(4 - x^2 - 4y^2 + 10y)$$

$$\frac{\partial Q_1}{\partial x} = \frac{2}{5}x, \quad \frac{\partial Q_1}{\partial y} = -\frac{1}{5},$$

$$\frac{\partial Q_2}{\partial x} = -\frac{1}{5}x, \quad \frac{\partial Q_2}{\partial y} = \frac{1}{10}(-8+10)$$

$\therefore$  At  $(x_0, y_0) = (0.4, 1.0)$  Apply condition

$$\Rightarrow \left| \frac{\partial Q_1}{\partial x} \right| + \left| \frac{\partial Q_1}{\partial y} \right| = \left| \frac{2}{5} \times 0.4 \right| + \left| \frac{1}{5} \right| = 0.36 < 1$$

$$\Rightarrow \left| \frac{\partial Q_2}{\partial x} \right| + \left| \frac{\partial Q_2}{\partial y} \right| = \left| -\frac{1}{5} \times 0.4 \right| + \left| \frac{1}{10}(2) \right| = 0.28 < 1$$

$\therefore$  The System is Converge.

Iteration : 0.1:-

put  $n=0$  in eq (i) & (ii) =

$$x_1 = \frac{1}{5}(x_0^2 - y_0 + 3) = \frac{1}{5}((0.4)^2 - (1) + 3) = 0.432$$

$$\therefore y_1 = \frac{1}{10}(4 - x_1^2 - 4y_0^2 + 10y_0) = \frac{1}{10}(4 - (0.432)^2 - 4 \times 1 + 10 \times 1)$$

$$= 0.081$$

$$\therefore x_1 = 0.432$$

$$y_1 = 0.081$$

# Interpolation

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## # Interpolation:-

if,

$p = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \rightarrow n^{\text{th}}$  degree polynomial

Let,

$(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  are  $n$  set of points

## Newton Interpolating polynomial

Formula:-

$$1. f(x) = p = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + \dots + a_n(x-x_0)(x-x_1)\dots(x-x_{n-1})$$

Example: 1:-

$\Rightarrow$  Find the polynomial of least degree which takes the values

|        |      |     |     |
|--------|------|-----|-----|
| $x$    | $-1$ | $1$ | $2$ |
| $f(x)$ | $0$  | $3$ | $6$ |

$\Rightarrow$  we know:-

$$f(x) = p = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1)$$

$$\therefore p = a_0 + a_1(x+1) + a_2(x+1)(x-1) \quad \text{--- (1)}$$

Now, For  $x = -1, f(x) = 0$ :-

$$0 = a_0 + a_1(-1+1) + a_2(-1+1)(-1-1)$$

$$\therefore a_0 = 0$$

For  $a_0 = 0, x = 1, f(x) = 3$ :-

$$\therefore 3 = a_0 + a_1(1+1) + a_2(1+1)(1-1)$$

$$\therefore a_1 = \frac{3 - a_0}{2} = \frac{3 - 0}{2}$$

$$\boxed{\therefore a_1 = -3}$$

For  $x = 2, f(x) = 6$ :-

$$\therefore 6 = a_0 + a_1(2+1) + a_2(2+1)(2-1)$$

$$\therefore a_2 = \frac{6 - 0 + 3(3)}{3} = \frac{6}{3}$$

$$\boxed{\therefore a_2 = 2}$$

Now put the values of  $a_0, a_1, a_2$  in eq (1) :-

$$\therefore P = 0 + (-3)(x+1) + 2(x+1)(x-1)$$

$$= 0 - 3x - 3 + 2x^2 - 2$$

$$\boxed{P = 2x^2 - 3x - 4}$$

Ans:-

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# Example 2:-

⇒ Find the polynomial of least degree which takes the values

|        |    |   |   |    |
|--------|----|---|---|----|
| $x$    | -1 | 1 | 2 | 5  |
| $f(x)$ | 0  | 3 | 6 | 39 |

⇒ we know:-

$$f(x) = p = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$$

$$\therefore f(x) = a_0 + a_1(x+1) + a_2(x+1)(x-1) + a_3(x+1)(x-1)(x-2) \quad \text{--- (1)}$$

Now, For  $x = -1, f(x) = 0$ :-

$$0 = a_0 + (-1+1)a_1 + a_2(-1+1)(-1-1) + a_3(-1+1)(-1-1)(-1-2)$$

$$\therefore a_0 = 0$$

For  $x = 1, f(x) = 3$ :-

$$3 = 0 + a_1(1+1) + a_2(1+1)(1-1) + a_3(1+1)(1-1)(1-2)$$

$$\therefore a_1 = \frac{3-0}{2} = \frac{3}{2}$$

For  $x = 2, f(x) = 6$ :-

$$6 = 0 + a_1(2+1) + a_2(2+1)(2-1) + a_3(2+1)(2-1)(2-2)$$

$$\therefore a_2 = \frac{6-0-3 \times \frac{3}{2}}{3} = 2$$

For  $x=5$ ,  $f(x)=30$ :-

$$\therefore 30 = 0 - 3(5+1) + 2(5+1)(5-1) + a_3(5+1)(5-1)(5-2)$$

$$\therefore a_3 = \frac{30 - 0 + 18 - 48}{72}$$

$$\therefore a_3 = 0$$

$\therefore$  put the value  $a_0, a_1, a_2, a_3$  in eq (1) :-

$$\begin{aligned} \therefore p &= 0 - 3x(x+1) + 2(x+1)(x-1) + 0 \\ &= 2x^2 - 3x - 4 \end{aligned}$$

Ans:-

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## # Lagrange Interpolation:-

Let  $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  are  $n$ -set of points  
the Lagrange interpolation is:-

Formula:-

$$1. \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} y_1$$

$$+ \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_2-x_0)(x_2-x_1)\dots(x_2-x_{n-1})} y_2 + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} y_n$$

# Example: 1:-

⇒ The following table gives the values as

|        |             |             |            |
|--------|-------------|-------------|------------|
| $x$    | $x_0$<br>0  | $x_1$<br>1  | $x_2$<br>2 |
| $f(x)$ | $y_0$<br>-4 | $y_1$<br>-1 | $y_2$<br>8 |

Using Lagrange interpolation formula

① find  $f(1.5)$

② find root of  $f(x) = 0$

⇒ we know:-

$$f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2$$

$$\therefore f(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} \times (-4) + \frac{(x-0)(x-2)}{(1-0)(1-2)} \times (-1) + \frac{(x-0)(x-1)}{(2-0)(2-1)} \times 8$$

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$$\therefore f(x) = \frac{(x-1)(x-2)}{2} \times (-4) + \frac{x(x-2)}{-1} \times (-1) + \frac{x(x-1) \times 8}{2} \quad \text{--- (1)}$$

(1)

$$\begin{aligned} \therefore f(1.5) &= \frac{(1.5-1)(1.5-2)}{2} \times (-4) + \frac{1.5 \times (1.5-2)}{-1} \times (-1) + \frac{1.5 \times (1.5-1) \times 8}{2} \\ &= \frac{1}{2} - \frac{3}{4} + 3 \\ &= 2.75 \end{aligned}$$

(11)

⇒ For root finding:—

Formula:—

$$\begin{aligned} \therefore x_0 &= \frac{(y-y_1)(y-y_2)\dots(y-y_n)}{(y_0-y_1)(y_0-y_2)\dots(y_0-y_n)} \times x_0 + \frac{(y-y_0)(y-y_2)\dots(y-y_n)}{(y_1-y_0)(y_1-y_2)\dots(y_1-y_n)} x_1 \\ &+ \frac{(y-y_0)(y-y_1)\dots(y-y_{n-1})}{(y_2-y_0)(y_2-y_1)\dots(y_2-y_{n-1})} x_2 + \dots + \frac{(y-y_0)(y-y_1)\dots(y-y_{n-1})}{(y_n-y_0)(y_n-y_1)\dots(y_n-y_{n-1})} x_n \end{aligned}$$

∴ we know:—

$$x = \frac{(y-y_1)(y-y_2)}{(y_0-y_1)(y_0-y_2)} \times x_0 + \frac{(y-y_0)(y-y_2)}{(y_1-y_0)(y_1-y_2)} x_1 + \frac{(y-y_0)(y-y_1)}{(y_2-y_0)(y_2-y_1)} x_2$$

$$\therefore x_0 = \frac{(y+1)(y-8)}{(-4+1)(-4-8)} \times 0 + \frac{(y+4)(y-8)}{(-1+4)(-1-8)} \times 1 + \frac{(y+4)(y+1)}{(8+4)(8+1)} \times 2$$

(1)

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for  $x=0$ :-

$$x = 0 + \frac{4x-8}{0x-0} \cdot x_1 + \frac{4x+1}{12x-0} \cdot x_2$$

$$\therefore x_0 = \frac{0x}{27} + \frac{2}{27}$$

$$\therefore x_0 = 1.26$$

# Example : 2

Consider the table

|        |    |    |     |     |      |
|--------|----|----|-----|-----|------|
| $x$    | -2 | 0  | 3   | 6   | 7    |
| $f(x)$ | 2  | -4 | -58 | 842 | 1802 |

using Lagrange interpolation formula

- ① find  $f(x)$
- ② find a real root of  $f(x) = 0$
- ③ Write down MATLAB codes using "polyfit( $x_0, y, n$ )" and "polyval( $p, x$ )" for  $n_0 = 1, 5, 0$

we know:-

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} \cdot y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} \cdot y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} \cdot y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} \cdot y_3 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} \cdot y_4 \quad \text{①}$$

$$\therefore f(x) = \frac{(x-0)(x-3)(x-6)(x-7)}{(-2-0)(-2-3)(-2-6)(-2-7)} x^2 + \frac{(x+2)(x-3)(x-6)(x-7)}{(0+2)(0-3)(0-6)(0-7)} x(-4) \\ + \frac{(x+2)(x-0)(x-6)(x-7)}{(3+2)(3-0)(3-6)(3-7)} x(-58) + \frac{(x+2)(x-0)(x-3)(x-7) \times 842}{(6+2)(6-0)(6-3)(6-7)} \\ + \frac{(x+2)(x-0)(x-3)(x-6) \times 1802}{(7+2)(7-0)(7-3)(7-6)} \quad \text{--- (11)}$$

$$\therefore f(4) = \frac{4 \times 1 \times -2 \times -3}{-2 \times -5 \times -8 \times -9} x^2 + \frac{6 \times 1 \times -2 \times -3 \times -4}{2 \times -3 \times -6 \times -7} + \\ \frac{6 \times 4 \times -2 \times -3 \times -58}{5 \times 3 \times -3 \times -4} + \frac{(6 \times 4) \times (1) \times (-3) \times 842}{8 \times 6 \times 3 \times -1} \\ + \frac{6 \times 4 \times -2 \times -3 \times 1802}{9 \times 7 \times 4 \times 1} \\ = \frac{1}{15} + \frac{4}{7} + \frac{232}{5} + 421 - \frac{7208}{24}$$

$$f(4) = 124.8$$

Ans:-

(11)

We know:-

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} x_0 + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} x_1 \\ + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} x_2 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} x_3 \\ + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} x_4$$

∴ For  $y=0$ :-

$$x_6 = \frac{+4 \times 58 \times (-842) \times 1802}{(2+4)(2+58)(2-842)(2-1802)} x^{-2} + \frac{(-2)(+58)(-842)(1802)(0)}{(-4-2)(-4+58)(-4-842)(-4-1802)}$$

$$+ \frac{(-2)(+4)(-842)(-1802)}{(-58-2)(-58+4)(-58-842)(-58-1802)} x^3 + \frac{-2x+4x+58x+1802x+6}{840 \times 846 \times 960 \times -960}$$

$$+ \frac{(-2) \times (+4) \times (+58) \times (-842) \times 7}{1800 \times 1806 \times 1860 \times 960}$$

$$\therefore x_6 = 0.65 \times (-2) + 0 + (-6.71 \times 10^{-3})$$

$$- 8.170 \times 10^{-6} + 4.71 \times 10^{-7}$$

$$\therefore x_6 = -1.306$$

⇒ MATLAB Code:-

>> clear;

>>  $x = [-2, 0, 3, 6, 7];$

>>  $y = [2, -4, -58, 842, 1802];$

>>  $p = \text{polyfit}(x, y, 4)$

>>  $x_1 = [2, 5, 9];$

>>  $y_1 = \text{polyval}(p, x_1)$

here,

$y = f(x)$

value of  $x$

$p = \text{polyfit}(x, y, 4)$

value of  $f(x)$

least degree  $n=4$

## # Divided Difference Interpolation:-

Let,  $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots$  some set of points.

Then

| $x$   | $f(x)$ | 1DD                                        | 2DD                                                | 3DD                                                |
|-------|--------|--------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| $x_0$ | $y_0$  |                                            |                                                    |                                                    |
| $x_1$ | $y_1$  | $= \frac{y_1 - y_0}{x_1 - x_0} = 1DD(y_1)$ |                                                    |                                                    |
| $x_2$ | $y_2$  | $= \frac{y_2 - y_1}{x_2 - x_1} = 1DD(y_2)$ | $2DD(y_0) = \frac{1DD(y_2) - 1DD(y_1)}{x_2 - x_0}$ |                                                    |
| $x_3$ | $y_3$  | $= \frac{y_3 - y_2}{x_3 - x_2}$            | $2DD(y_2) = \frac{1DD(y_3) - 1DD(y_2)}{x_3 - x_1}$ | $3DD(y_0) = \frac{2DD(y_3) - 2DD(y_2)}{x_3 - x_0}$ |

Formula:-

$$(1) f(x) = y_0 + 1DD(x - x_0) + 2DD(x - x_0)(x - x_1) + 3DD(x - x_0)(x - x_1)(x - x_2) + \dots + nDD(x - x_0)(x - x_1) \dots (x - x_{n-1})$$

$$(2) g(x) = f(x) + b(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)$$

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# Example 1.1 —

⇒ Consider

|        |    |    |   |    |
|--------|----|----|---|----|
| $x$    | -1 | 1  | 2 | 3  |
| $f(x)$ | -7 | -1 | 8 | 20 |

- (i) Construct a divided-difference table using above data
- (ii) Find the least degree polynomial and find  $f(u)$
- (iii) Find a real root of  $f(x) = 0$
- (iv) Find polynomial  $g(x)$  using values from table and  $g(4) = 103$
- (v) Write MATLAB code using "polyfit(m,y,n)", and "polyval(p,x)" for  $x = 1.5, 2.5, 3.5$ .

⇒

①

| $x$ | $f(x)$ | $1^{st} \text{ D.D.}$     | $2^{nd} \text{ D.D.}$      | $3^{rd} \text{ D.D.}$         |
|-----|--------|---------------------------|----------------------------|-------------------------------|
| -1  | -7     |                           |                            |                               |
| 1   | -1     | $= \frac{-1+7}{1+1} = 3$  |                            |                               |
| 2   | 8      | $= \frac{8+1}{2-1} = 9$   | $= \frac{9-3}{2+1} = 2$    |                               |
| 3   | 20     | $= \frac{20-8}{3-2} = 12$ | $= \frac{12-9}{3-1} = 1.5$ | $= \frac{2-1.5}{3+1} = 0.125$ |

(ii)

we know:-

The formula for least degree polynomial is

$$\Rightarrow f(x) = y_0 + 1DD(x-x_0) + 2DD(x-x_0)(x-x_1) + 3DD(x-x_0)(x-x_1)(x-x_2)$$

$$\begin{aligned} \therefore f(x) &= -7 + 3(x-x_0) + 2(x-x_0)(x-x_1) + 1(x-x_0)(x-x_1)(x-x_2) \\ &= -7 + 3x(x+1) + 2(x+1)(x-1) + (x+1)(x-1)(x-2) \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} \therefore f(4) &= -7 + 3x(4+1) + 2(4+1)(4-1) + (4+1)(4-1)(4-2) \\ &= -7 + 15 + 30 + 30 \\ &= 68 \end{aligned}$$

(iii)

 $\Rightarrow$  To find the root we have to check the  $f(a) \cdot f(b) < 0$ For (-1, 1):-

$$\therefore f(-1) \times f(1) = (-7) \times (-1) = 7 > 0$$

For (1, 2):-

$$f(1) \times f(2) = (-1) \times (8) = -8 < 0$$

For (2, 3):-

$$f(2) \times f(3) = (8) \times (29) = 236 > 0$$

$\therefore$  The root of  $f(x)$  is lies in  $(1, 2)$

$\Rightarrow$  Now

| $x$ | $f(x)$ | $\Delta DD$             |
|-----|--------|-------------------------|
| 1   | -1     |                         |
| 2   | 8      | $= \frac{8+1}{2-1} = 9$ |

$$\therefore f(x) = y_0 + \Delta DD (x - x_0)$$

$$0 = -1 + 9(x - 1)$$

$$0 = -1 + 9x - 9$$

$$\therefore x = \frac{10}{9} = 1.1$$

$\therefore x = 1.1$  is a root of  $f(x) = 0$

(iv)

we know:-

$$g(x) = f(x) + b(x - x_0)(x - x_1)(x - x_2)(x - x_3)$$

$$\therefore g(x) = 68 + b(x+1)(x-1)(x-2)(x-3) \quad \text{--- (1)}$$

$$\therefore g(4) = 68 + b(4+1)(4-1)(4-2)(4-3)$$

$$103 = 68 + b(4+1)(4-1)(4-2)(4-3)$$

$$\therefore b = \frac{103-68}{30} = 1.167$$

put the value of  $b$  in eq (1) =

$$\therefore g(x) = f(x) + 1.1667(x+1)(x-1)(x-2)(x-3)$$

Ans:-

(V)

⇒ MATLAB CODE:-

>> clear;

>> x = [-1, 1, 2, 3];

>> y = [-7, -1, 8, 20];

>> p = polyfit(x, y, 3);

>> x1 = [1.5, 2.5, 3.5];

>> y1 = polyval(p, x1)

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# Exercise : 5.1

① The table below gives the velocity  $v$  at time  $t$

|          |   |   |    |     |
|----------|---|---|----|-----|
| $t(s)$   | 1 | 3 | 4  | 7   |
| $v(m/s)$ | 3 | 5 | 21 | 201 |

i Construct a divided-difference table for the above data.

ii Find the least degree polynomial.

iii Find the acceleration at time  $t = 6s$ .

iv Find the distance function when  $s(0) = 2$ .

⇒

①

| $t(s)$ | $v(m/s)$ | 1DD                         | 2DD                        | 3DD                      |
|--------|----------|-----------------------------|----------------------------|--------------------------|
| 1      | 3        |                             |                            |                          |
| 3      | 5        | $= \frac{5-3}{3-1} = 1$     |                            |                          |
| 4      | 21       | $= \frac{21-5}{4-3} = 16$   | $= \frac{16-1}{4-1} = 5$   |                          |
| 7      | 201      | $= \frac{201-21}{7-4} = 60$ | $= \frac{60-16}{7-3} = 11$ | $= \frac{11-5}{7-1} = 1$ |

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(ii)

We know:-

$$f(x) = y_0 + 1DD(x-x_0) + 2DD(x-x_0)(x-x_1) + 3DD(x-x_0)(x-x_1)(x-x_2)$$

$$\therefore f(x) = 3 + 1x(x-1) + 5(x-1)(x-3) + 1(x-1)(x-3)(x-4)$$

$\therefore$  The least degree polynomial is:-

$$f(x) = 3 + (x-1) + 5(x-1)(x-3) + (x-1)(x-3)(x-4)$$

Ans:-

here  
 $f(x) = v$

(iii)

$$\begin{aligned} \therefore f(6) &= 3 + (6-1) + 5(6-1)(6-3) + (6-1)(6-3)(6-4) \\ &= 3 + 5 + 75 + 30 \\ &= 113 \end{aligned}$$

$\therefore$  The acceleration is  $113 \text{ ms}^{-1}$  at  $t = 6 \text{ s}$ .

(iv)

We know:-

$$g(x) = f(x) + b(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$$\therefore g(0) = f(0) + b(0-1)(0-3)(0-4)(0-7) \quad \text{--- (1)}$$

$$\begin{aligned} \therefore 2 &= 3 + 1x-1 + 5x-1x-3 + 1x-1x-3x-4 + 84b \\ b &= \frac{2-5}{84} = -\frac{1}{28} \end{aligned}$$

here  
 $g(x) = s(x)$   
 $f(x) = v$

$\therefore$  put the value in eq (1) =

$$\therefore S(x) = f(x) = f(n) = 15 - \frac{1}{28} (x-1)(x-2)(x-4)(x-7)$$

Ans:-

(4) The table below gives the values  $x$  and  $f(x)$

|        |    |    |    |    |
|--------|----|----|----|----|
| $x$    | -2 | -1 | 0  | 3  |
| $f(x)$ | 12 | 14 | 10 | 22 |

- (i) Construct a D.D table for the above data.
- (ii) Find the least degree polynomial and find  $f(8)$
- (iii) Given  $g(8) = 1202$ , find the polynomial  $g(x)$  that also takes the values of the above table.
- (iv) Use Lagrange interpolation formula to find
  - (a) a real root of  $f(x) = 0$  using linear approximation.
  - (b) a real root of  $f(x) = 0$  using all points.
- (v) Write down MATLAB codes using "polyfit(x,y,n)" and "polyval(p,x)" for  $x = 1.5, 2.5, 3.5$

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(i)

| $x$ | $f(x)$ | $1DD$                    | $2DD$                   | $3DD$                 |
|-----|--------|--------------------------|-------------------------|-----------------------|
| -2  | 12     |                          |                         |                       |
| -1  | 14     | $\frac{14-12}{-1+2} = 2$ |                         |                       |
| 0   | 10     | $\frac{10-14}{0+1} = -4$ | $\frac{-4-2}{0+2} = -3$ |                       |
| 3   | 22     | $\frac{22-10}{3-0} = 4$  | $\frac{4+4}{3+1} = 2$   | $\frac{2+3}{3+2} = 1$ |

(ii)

We know:-

$$f(x) = y_0 + 1DD(x-x_0) + 2DD(x-x_0)(x-x_1) + 3DD(x-x_0)(x-x_1)(x-x_2)$$

$$\therefore f(x) = 12 + 2(x+2) - 3(x+2)(x+1) + 1 \times (x+2)(x+1)(x-0) \quad \text{--- (1)}$$

$$\begin{aligned} \therefore f(8) &= 12 + 2 \times (8+2) - 3(8+2)(8+1) + (8+2)(8+1)(8-0) \\ &= 12 + 20 - 270 + 720 \\ &= 582 \end{aligned}$$

Ans:-

iii

We know:-

$$g(x) = f(x) + b(x - x_0)(x - x_1)(x - x_2)(x - x_3) \quad \text{--- (1)}$$

$$\therefore g(8) = f(8) + b(8+2)(8+1)(8-0)(8-3)$$

$$\therefore 1202 = 582 + 3600b$$

$$\therefore b = \frac{1202 - 582}{3600}$$

$$b = 0.172$$

$\therefore$  put the value of  $b$  in eq (1)  $\Rightarrow$

$$\therefore g(x) = f(x) + 0.172(x+2)(x+1)(x-0)(x-3)$$

Ans:-

iv

(a) Linear Approximation:-

$\Rightarrow$  To find the root we have to check the condition of root finding  $f(a) \times f(b) < 0$ .

1. For  $(-2, -1)$ :-

$$f(-2) \times f(-1) = 12 \times 14 = 168 > 0$$

2. For  $(-1, 0)$ :-

$$f(-1) \times f(0) = 14 \times 10 = 140 > 0$$

4. For (0,3):-

$$f(0) \times f(3) = 10 \times 22 = 220 > 0$$

Since, No points meets the Condition,

So. The real root is not between this four point.

(b) Using all the points:-

⇒ We know:-

$$\begin{aligned} x &= \frac{(y-y_1)(y-y_2)(y-y_3)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)} x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)} x_1 + \\ &\quad \frac{(y-y_0)(y-y_1)(y-y_3)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)} x_2 + \frac{(y-y_0)(y-y_1)(y-y_2)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)} x_3 \\ &= \frac{(y-14)(y-10)(y-22)}{-2 \times 2 \times -10} x_{-2} + \frac{(y-12)(y-10)(y-22)}{2 \times 4 \times -8} x_{-1} \\ &\quad + \frac{(y-10)(y-14)(y-22)}{-2 \times -4 \times -12} x_0 + \frac{(y-12)(y-14)(y-10)}{10 \times 8 \times 12} x_3 \end{aligned}$$

For  $y=0$ :-

$$\begin{aligned} x_0 &= \frac{(-14) \times (-10) \times (-22)}{40} x_{-2} + \frac{(-12) \times (-10) \times (-22)}{2 \times 4 \times -8} x_{-1} + \frac{-12 \times -14 \times -10}{10 \times 8 \times 12} x_3 \\ &= 154 - 41.25 - 5.25 \\ &= 107.5 \end{aligned}$$

Ans:-

(V)

MATLAB Code:-

```

>> clear;
>> x = [-2, -1, 0, 3];
>> y = [12, 14, 10, 22];
>> p = polyfit(x, y, 3);
>> x1 = [1.5, 2.5, 3.5];
>> y1 = polyval(p, x1);

```

3. The table gives the values of  $x$  and  $f(x)$

|        |    |    |     |     |     |
|--------|----|----|-----|-----|-----|
| $x$    | 4  | 5  | 7   | 9   | 11  |
| $f(x)$ | 62 | 95 | 185 | 307 | 461 |

- (i) Construct a D.D table.
- (ii) Find the least degree polynomial. also find  $f(12)$
- (iii) Find the polynomial  $g(x)$  that takes the value of the above table. and  $g(12) = 1280$ .
- (iv) Using Lagrange interpolating to
  - (a) find the  $f(8)$  using 3 points.
  - (b) The value of  $x$  for  $f(x) = 380$  using three points.
- (v) Write down the MATLAB code using "polyfit(x, y, n)" and "polyval(p, x)" for  $x = 1, 3, 5$

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(i)

| $x$ | $f(x)$ | 100  | 200 | 300                      | 400 |
|-----|--------|------|-----|--------------------------|-----|
| 4   | 62     |      |     |                          |     |
| 5   | 05     | = 33 |     |                          |     |
| 7   | 185    | = 45 | = 4 |                          |     |
| 9   | 302    | = 61 | = 4 | = $\frac{4-4}{9-4} = 0$  |     |
| 11  | 461    | = 77 | = 4 | = $\frac{4-4}{11-4} = 0$ | = 0 |

(ii)

we know—

$$f(x) = f_0 + 100(x - x_0) + 200(x - x_0)(x - x_1) + \dots$$

$$\therefore f(x) = 62 + 33(x - 4) + 4(x - 4)(x - 5) + 0 \quad \text{--- (1)}$$

$$\therefore f(12) = 62 + 33(12 - 4) + 4(12 - 4)(12 - 5)$$

$$= 62 + 396 + 224$$

$$= 678$$

Ans.

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(iii)

we know:-

$$g(x) = f(x) + b(x-x_0)(x-x_1)(x-x_2)(x-x_3)(x-x_4)$$

$$\therefore g(x) = f(x) + b(x-4)(x-5)(x-7)(x-0)(x-11) \quad \text{--- (1)}$$

$$\therefore g(12) = f(12) + b(12-4)(12-5)(12-7)(12-0)(12-11)$$

$$1280 = 678 + 1080b$$

$$\therefore b = \frac{1280-678}{1080}$$

$$= 0.557$$

put the value of b in eq (1) =

$$\therefore g(x) = f(x) = 0.557(x-4)(x-5)(x-7)(x-0)(x-11)$$

Ans:-

(iv)

(a)

we know:-

$$f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} \times y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} \times y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} \times y_2$$

$$\therefore f(x) = \frac{(x-7)(x-5)}{(4-7)(4-5)} \times 62 + \frac{(x-4)(x-7)}{(5-4)(5-7)} \times 05 + \frac{(x-4)(x-5)}{(7-4)(7-5)} \times 185$$

$$= \frac{(x-7)(x-5)}{03} \times 62 + \frac{(x-4)(x-7)}{-2} \times 05 + \frac{(x-4)(x-5)}{6} \times 185$$

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$$\therefore f(8) = \frac{1 \times 3}{3} \times 62 + \frac{4 \times 1}{-2} \times 115 + \frac{4 \times 3}{6} \times 185$$

$$= 62 - 110 + 127$$

$$= 79$$

Ans:-

(b)

We know:-

$$x = \frac{(y-y_1)(y-y_2)}{(y_0-y_1)(y_0-y_2)} x_0 + \frac{(y-y_0)(y-y_2)}{(y_1-y_0)(y_1-y_2)} x_1 + \frac{(y-y_1)(y-y_0)}{(y_2-y_1)(y_2-y_0)} x_2$$

$$\therefore x_6 = \frac{(y-115)(y-185)}{(62-115)(62-185)} \times 4 + \frac{(y-62)(y-185)}{(115-62)(115-185)} \times 5 + \frac{(y-62)(y-115)}{(185-62)(185-115)} \times 7$$

For  $y = 130$ :-

$$\therefore x_6 = 54.77 + (-104.40) + 57.30$$

$$= 7.67$$

(✓)

MATLAB CODE:-

```
>> clear;
>> x=[4,7,11];
>> y=[62,115,185];
>> p=polyfit(x,y,4);
>> x1=[1,3,5];
>> y1=polyval(p,x1)
```

Q:- Find the polynomial of least degree which take the values

|        |    |   |    |
|--------|----|---|----|
| $x$    | 4  | 6 | 8  |
| $f(x)$ | 12 | 4 | 17 |

$\Rightarrow$  we know:-

$$f(x) = p = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$$

$$\therefore f(x) = a_0 + a_1(x - 4) + a_2(x - 4)(x - 6) \quad \text{--- (1)}$$

Now, For  $x=4$ ,  $f(x)=12$ :-

$$12 = a_0 + a_1(4-4) + a_2(4-4)(4-6)$$

$$\therefore a_0 = 12$$

For  $x=6$ ,  $f(x)=4$ :-

$$4 = 12 + a_1(6-4)$$

$$\therefore a_1 = \frac{4-12}{2}$$

$$\therefore a_1 = -2$$

For  $x=8$ ,  $f(x)=17$ :-

$$17 = 12 + (-2)(8-4) + a_2(8-4)(8-6)$$

$$\therefore a_2 = \frac{17-12+12}{8}$$

$$\therefore a_2 = 2.125$$

put the value of  $a_0, a_1, a_2$  in eq (1) =

$$\therefore f(x) = p = 12 + (-3)(x-4) + 2.12.5(x-4)(x-6)$$

Ans:-

Q:- Consider

|        |    |    |    |    |
|--------|----|----|----|----|
| $x$    | -2 | -1 | 1  | 2  |
| $f(x)$ | 6  | 7  | -8 | +2 |

$\Rightarrow$  Using D.D find the real root of  $f(x) = 0$

$\Rightarrow$  To find the root we have to check the condition  $f(a) \times f(b) < 0$

Now For (-2, -1) :-

$$\therefore f(-2) \times f(-1) = 6 \times 7 = 42 > 0$$

For (-1, 1) :-

$$\therefore f(-1) \times f(1) = 7 \times -8 = -56 < 0$$

For (1, 2) :-

$$f(1) \times f(2) = -8 \times 2 = -16 < 0$$

$\therefore$  The root of  $f(x)$  lies in  $(-1, 1)$  and  $(1, 2)$

$\Rightarrow$  Now,

| $x$ | $f(x)$ | ADD                         | 2DD                           |
|-----|--------|-----------------------------|-------------------------------|
| -1  | 7      |                             |                               |
| 1   | -8     | $= \frac{-8-7}{1+1} = -7.5$ |                               |
| 2   | 2      | $= \frac{2+8}{2-1} = 10$    | $= \frac{10+7.5}{0.5} = 5.83$ |

$$\therefore f(x) = y_0 + \text{ADD}(x+1) + 2\text{DD}(x+1)(x-1)$$

$$0 = 7 - 7.5(x+1) + 5.83(x+1)(x-1)$$

$$\Rightarrow 0 = 7 - 7.5x - 7.5 + 5.83(x^2 - 1)$$

$$\Rightarrow 0 = 7 - 7.5x - 7.5 + 5.83x^2 - 5.83$$

$$\Rightarrow 0 = -7.5x + 5.83x^2 - 6.33$$

$$\therefore 5.83x^2 - 7.5x - 6.33 = 0$$

$$\therefore x = 1.87, -0.582$$

where,

$x = -0.582$  is lies in  $(-1, 1)$  And

$x = 1.87$  is lies in  $(1, 2)$

Both is a root of  $f(x) = 0$

Ans:-

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# Spline Interpolation

Chaos

## # Spline Interpolation:-

### # Formula:-

$$\underline{\text{1.}} \quad f_k(x) = y_{k-1} + \frac{\Delta y_k}{h_k} (x - x_{k-1}) \quad ; \quad (k=1, 2, \dots, n-1)$$

$$* \Delta y_k = y_k - y_{k-1}$$

$$* h_k = x_k - x_{k-1}$$

So, the function  $f(x)$  of the form:-

$$f(x) = \begin{cases} f_1(x), & x_0 \leq x < x_1 \\ f_2(x), & x_1 \leq x < x_2 \\ \vdots & \\ f_{n-1}(x), & x_{n-2} \leq x < x_{n-1} \end{cases}$$

### # Example: 6.1:-

⇒ Find the linear spline for the following data set

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| $x$ | -1  | 1   | 2   | 5   |
| $y$ | 2.2 | 9.5 | 5.4 | 1.5 |

Hence, estimate the value of  $y(1.5)$ .

⇒ We know:-

$$f_k(x) = y_{k-1} + \frac{y_k - y_{k-1}}{x_k - x_{k-1}} (x - x_{k-1})$$

$$\therefore f_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$$

$$\therefore f_1(x) = 2.2 + \frac{3.5 - 2.2}{1 - (-1)} \times (x - (-1))$$

$$= 2.2 + \frac{1.3}{2} \times (x + 1)$$

$$-1 \leq x \leq 1$$

$$f_2(x) = y_{2-1} + \frac{y_2 - y_1}{x_2 - x_1} \times (x - x_1)$$

$$= y_1 + \frac{5.4 - 3.5}{2 - 1} (x - x_1)$$

$$= 3.5 + \frac{1.9}{1} (x - 1)$$

$$1 \leq x \leq 2$$

$$f_3(x) = y_{3-1} + \frac{y_3 - y_2}{x_3 - x_2} \times (x - x_{3-1})$$

$$= y_2 + \frac{y_3 - y_2}{x_3 - x_2} \times (x - x_2)$$

$$= 5.4 + \frac{1.5 - 5.4}{5 - 2} (x - 2)$$

$$= 5.4 + \frac{-3.9}{3} (x - 2)$$

$$2 \leq x \leq 5$$

So, the linear spline function is:-

$$f(x) = \begin{cases} 2.2 + 0.65(x + 1), & -1 \leq x \leq 1 \\ 3.5 + 1.9(x - 1), & 1 \leq x \leq 2 \\ 5.4 - 1.3(x - 2), & 2 \leq x \leq 5 \end{cases}$$

At  $x = 1.5$  is in  $1 \leq x \leq 2$

$$\therefore f(1.5) = 3.5 + 1.9(1.5 - 1)$$

$$= 3.5 + 1.9 \times 0.5$$

$$= 4.45$$

## # Linear spline Interpolation:-

Formula:-

$$\hat{f}_k(x) = y_k + \frac{\Delta y_k}{h_k} (x - x_k) ; (k = 1, 2, \dots, n-1)$$

$$* \Delta y_k = y_k - y_{k-1}$$

$$* h_k = x_k - x_{k-1}$$

So, the function  $f(x)$  is:-

$$f(x) = \begin{cases} f_1(x), & x_0 \leq x \leq x_1 \\ f_2(x), & x_1 \leq x \leq x_2 \\ \vdots & \\ f_{n-1}(x), & x_{n-2} \leq x \leq x_{n-1} \end{cases}$$

# Example:- 6.1:-

⇒ Find the linear spline for the following data set

|   |     |     |     |     |
|---|-----|-----|-----|-----|
| X | -1  | 1   | 2   | 5   |
| Y | 2.2 | 3.5 | 5.4 | 1.5 |

Hence, Estimate the value of (1.5)

⇒ we know:-

$$f_k(x) = y_k + \frac{\Delta y_k}{h_k} (x - x_k)$$

$$\therefore f_1(x) = y_1 + \frac{y_1 - y_0}{x_1 - x_0} \times (x - x_0)$$

$$= 3.5 + \frac{3.5 - 2.2}{1 - (-1)} \times (x - (-1))$$

$$-1 \leq x \leq 1$$

$$\therefore f_1(x) = 3.5 + \frac{1.3}{2} (x-1)$$

$$f_2(x) = 5.4 + \frac{5.4-3.5}{2-1} (x-2)$$

$$= 5.4 + \frac{5.4-3.5}{2-1} (x-2)$$

$$1 \leq x \leq 2$$

$$= 5.4 + 1.9(x-2)$$

$$f_3(x) = 1.5 + \frac{1.5-5.4}{5-2} (x-5)$$

$$= 1.5 + \frac{1.5-5.4}{5-2} (x-5)$$

$$2 \leq x \leq 5$$

$$= 1.5 - 1.3(x-5)$$

So, the Linear spline function is:—

$$f(x) = \begin{cases} 3.5 + 0.65(x-1) & ; 1 \leq x \leq 2 \\ 5.4 + 1.9(x-2) & ; 2 \leq x \leq 5 \\ 1.5 - 1.3(x-5) & ; 5 \leq x \leq 7 \end{cases}$$

At  $x=1.5$  is in  $1 \leq x \leq 2$

$$\therefore f(1.5) = 5.4 + 1.9(1.5-2)$$

$$= 4.45$$

## ## Cubic Spline Interpolation:-

⇒ There are 3 types:-

(i) Natural.

(ii) Clamped

(iii) Not-a-knot.

⇒ The Condition of Natural Cubic spline:-

$$(*) f''_1(x_0) = 0$$

$$(*) f''_n(x_n) = 0$$

⇒ The Condition of Clamped Cubic spline:-

$$(*) f'_1(x_0) = d_0$$

$$(*) f'_n(x_n) = d_n$$

# Example: 6.2

⇒ A natural cubic spline is defined by

$$f(x) = \begin{cases} A(x+1)^3 - B(x+1)^2 + C(x+1) + 1; & (-1 \leq x \leq 1) \\ D(x-1)^3 + 6(x-1)^2 + E(x-1)(-1); & (1 \leq x \leq 2) \end{cases}$$

(i) Use continuity and boundary conditions to estimate A, B, C, D and E.

(ii) Find the value of  $f(1.4)$  from the spline curve.

(iii) Write a MATLAB code using build-in function "sp = spline(x,y)" and "fval(sp,x)" which interpolate data  $(-1, -3)$ ,  $(1, 6)$ ,  $(2, 12)$  and find  $f(x)$  for  $x = 1.1, 2.2$ .

①

⇒ In given:-

$$f(x) = \begin{cases} A(x+1)^3 - B(x+1)^2 + C(x+1) + 1; & (-1 \leq x \leq 1) \\ D(x-1)^3 + 6(x-1)^2 + E(x-1)(-1); & (1 \leq x \leq 2) \end{cases} \quad \text{--- ①}$$

Let,

$$f_1(x) = A(x+1)^3 - B(x+1)^2 + C(x+1) + 1 \quad \text{--- ②}$$

$$f_2(x) = D(x-1)^3 + 6(x-1)^2 + E(x-1)(-1) \quad \text{--- ③}$$

Now 1st and 2nd derivatives of (ii) & (iii) becomes:-

$$f'_1(x) = 3A(x+1)^2 + 2B(x+1) + C$$

$$f'_2(x) = 3D(x-1)^2 + 6 \times 2(x-1) + E$$

sol  $f''_1(x) = 6A(x+1) + 2B$

$$f''_2(x) = 6D(x-1) + 12$$

At  $x=1$  → Common point

| $f_1(x) = f_2(x)$                                                                                                  | $f'_1(x) = f'_2(x)$                     | $f''_1(x) = f''_2(x)$                                             |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------|
| $8A - 4B + 2C + 1 = -1$<br>$\Rightarrow 8A - 4B + 2C = -2$<br>$\Rightarrow 4A - 2B + C = -1 \quad \text{--- (iv)}$ | $12A + 4D + C = E \quad \text{--- (v)}$ | $12A + 2B = 12$<br>$\Rightarrow 6A + B = 6 \quad \text{--- (vi)}$ |

For Natural cubic spline:-

We know:-

$$f''_1(x_0) = 0 \quad \& \quad f''_2(x_2) = 0$$

$$1. \quad f''_1(-1) = 0 \quad \& \quad f''_2(2) = 0$$

$$\therefore f''_1(-1) = 6A(-1+1) + 2B = 0 \quad \& \quad f''_2(2) = 6D(2-1) + 12 = 0$$

$$\Rightarrow 2B = 0$$

$$\boxed{B = 0}$$

$$\Rightarrow 6D = -12$$

$$\Rightarrow \boxed{D = -2}$$

$\therefore$  put the value of  $B$  in eq (vi)  $\Rightarrow$

$$6A + B = 6$$

$$\therefore \boxed{A = \frac{6}{6} = 1}$$

put the value of  $A, B$  in eq (iv)  $\Rightarrow$

$$4A + 2B + C = 1$$

$$\therefore \boxed{C = -3 - 2}$$

$$\therefore \boxed{C = -5}$$

put the value of  $A, B, C$  in eq (v)  $\Rightarrow$

$$12A + 4B + C = E$$

$$\therefore E = 12 - 3 - 2$$

$$\therefore \boxed{E = 7}$$

Now, put the all value in eq (i)  $\Rightarrow$

$\therefore$  The natural cubic spline is: -

$$f(x) = \begin{cases} (x+1)^3 + 0 \cdot (x+1)^2 - 5(x+1) + 1; & -1 \leq x \leq 1 \\ -2(x-1)^3 + 6(x-1)^2 + 7(x-1) - 1; & 1 \leq x \leq 2 \end{cases}$$

Ans: -

ii

$x = 1.4$  lies inside  $1 \leq x \leq 2$

Thus  $f_2(x) = -2(x-1)^3 + 6(x-1)^2 + 7(x-1) - 1$

$$\therefore f(1.4) = -2(1.4-1)^3 + 6(1.4-1)^2 + 7(1.4-1) - 1$$

$$= 2.632$$

iii

⇒ MATLAB CODE:-

```
>> clear;
>> x = [-1, 1, 2];
>> y = [-9, 6, 12];
>> sp = spline(x, y);
>> x1 = [1.1, 2.2];
>> y1 = fval(sp, x1)
```

here,

$(x, y) = (-1, -9)$   
 $(x, y) = (1, 6)$   
 $(x, y) = (2, 12)$

# problem 1:-

⇒ In the previous question, given that  $f'(1) = 2$  and  $f(2) = 6$

(i) use continuity and boundary conditions to find the values of A, B, C, D, E

(ii) Estimate  $f(0.5)$

(ii)

 $x = 1.4$  lies inside  $1 \leq x \leq 2$ 

$$\text{Thus } f_2(x) = -2(x-1)^3 + 6(x-1)^2 + 7(x-1) - 1$$

$$\therefore f(1.4) = -2(1.4-1)^3 + 6(1.4-1)^2 + 7(1.4-1) - 1$$

$$= 2.632$$

(iii)

⇒ MATLAB CODE:-

```

>> clear;
>> x = [-1, 1, 2];
>> y = [-9, 6, 12];
>> sp = spline(x, y);
>> x1 = [1.1, 2.2];
>> y1 = fval(sp, x1);

```

here,

$$(x, y) = (-1, -9)$$

$$(x, y) = (1, 6)$$

$$(x, y) = (2, 12)$$

# problem 1:-

⇒ In the previous question, given that  $f'(-1) = 2$  and  $f(2) = 6$

(i) Use continuity and boundary conditions to find the values of A, B, C, D, E

(ii) Estimate  $f(0.5)$

(i)

Is given:-

$$f_1(x) = \begin{cases} A(x+1)^3 + B(x+1)^2 + C(x+1) + 1; & -1 < x < 1 \\ D(x-1)^3 + 6(x-1)^2 + E(x-1) - 1; & 1 \leq x \leq 2 \end{cases} \quad \text{--- (1)}$$

Let

$$f_1(x) = A(x+1)^3 + B(x+1)^2 + C(x+1) + 1 \quad \text{--- (ii)}$$

$$f_2(x) = D(x-1)^3 + 6(x-1)^2 + E(x-1) - 1 \quad \text{--- (iii)}$$

Now 1st derivatives of (ii) & (iii) becomes:-

$$f_1'(x) = 3A(x+1)^2 + 2B(x+1) + C$$

$$f_2'(x) = 3D(x-1)^2 + 12(x-1) + E$$

and 2nd derivatives of (ii) & (iii) becomes:-

$$f_1''(x) = 6A(x+1) + 2B$$

$$f_2''(x) = 6D(x-1) + 12$$

At common point  $x=1$ :-

| $f_1(x) = f_2(x)$                                    | $f_1'(x) = f_2'(x)$                     | $f_1''(x) = f_2''(x)$                           |
|------------------------------------------------------|-----------------------------------------|-------------------------------------------------|
| $8A + 4B + 2C + 1 = -1$                              | $12A + 4B + C = E \quad \text{--- (v)}$ | $12A + 2B = 12$                                 |
| $\Rightarrow 8A + 4B + 2C = -2$                      |                                         | $\Rightarrow 6A + B = 6 \quad \text{--- (vi)}$  |
| $\Rightarrow 4A + 2B + C = -1 \quad \text{--- (iv)}$ |                                         | $\Rightarrow B = 6 - 6A \quad \text{--- (vii)}$ |

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⇒ For the clamped cubic spline:-

⇒ we know:-

$$f_1'(a_0) = d_0 \quad \& \quad f_2'(a_2) = d_2$$

$$\therefore f_1'(-1) = 2 \quad \& \quad f_2'(2) = 6$$

$$\Rightarrow 3A(-1+1)^2 + 2B(-1+1) + C = 2 \quad \& \quad \Rightarrow 3D(2-1)^2 + 12(-1+2) + E = 6$$

$$\boxed{\therefore C = 2}$$

$$\Rightarrow 3D + 12 + E = 6$$

$$\Rightarrow 3D + E = -6 \quad \text{--- (viii)}$$

put the value of C and B in eq (iv) :-

$$\Rightarrow 4A + 2B + C = -1$$

$$\Rightarrow 4A + 2 \times (6 - 6A) + 2 = -1$$

$$\Rightarrow 4A + 12 - 12A + 2 = -1$$

$$\Rightarrow -8A = -1 - 14$$

$$\therefore \boxed{A = \frac{-15}{-8} = \frac{15}{8}}$$

put the value of A in eq (vii) :-

$$B = 6 - 6A$$

$$= 6 - 6 \times \frac{15}{8}$$

$$\boxed{B = -\frac{21}{4}}$$

put the value of A, B, C in eq (V) =

$$\therefore E = 12A + 4B + C$$

$$= 12 \times \frac{15}{8} + 4 \times -\frac{21}{4} + 2$$

$$\therefore E = \frac{7}{2}$$

put the value of E in eq (VIII) =

$$3D + E = -6$$

$$3D = -6 - E$$

$$\therefore D = \frac{-6 - E}{3} = \frac{-6 - \frac{7}{2}}{3} = -\frac{19}{6}$$

Now, put the all value in eq (I) =

$\therefore$  The clamped cubic spline:—

$$f(x) = \begin{cases} \frac{15}{8}(x+1)^3 + \frac{21}{4}(x+1)^2 + 2(x+1) + 1; & -1 \leq x \leq 1 \\ -\frac{19}{6}(x-1)^3 + 6(x-1)^2 + \frac{7}{2}(x-1) - 1; & 1 \leq x \leq 2 \end{cases}$$

Ans:—

(ii)

$x = 0.5$  lies in  $-1 \leq x \leq 1$

Thus,  $f_1(0.5) = \frac{15}{8}(0.5+1)^3 + \frac{21}{4}(0.5+1)^2 + 2(0.5+1) + 1$   
 $= 22.1406$

Ans:—

# Exercise 16.1 :-

① Find the linear spline for the following data set.

|     |      |      |      |      |
|-----|------|------|------|------|
| $x$ | 1.0  | 1.5  | 2.0  | 2.5  |
| $y$ | 0.24 | 0.27 | 0.31 | 0.36 |

hence, estimate the value of  $f(2.25)$

⇒ we know :-

$$f_k(x) = y_k + \frac{y_k - y_{k-1}}{x_k - x_{k-1}} (x - x_{k-1})$$

$$\begin{aligned} \therefore f_1(x) &= y_1 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0) \\ &= 0.27 + \frac{0.27 - 0.24}{1.5 - 1.0} (x - 1.5) \quad 1.0 \leq x \leq 1.5 \end{aligned}$$

$$\begin{aligned} f_2(x) &= y_2 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \\ &= 0.31 + \frac{0.31 - 0.27}{2.0 - 1.5} (x - 2.0) \quad 1.5 \leq x \leq 2.0 \end{aligned}$$

$$\begin{aligned} f_3(x) &= y_3 + \frac{y_3 - y_2}{x_3 - x_2} (x - x_2) \\ &= 0.36 + \frac{0.36 - 0.31}{2.5 - 2.0} (x - 2.5) \quad 2.0 \leq x \leq 2.5 \end{aligned}$$

∴ The linear Spline function is:-

$$f(x) = \begin{cases} 0.27 + 0.06(x - 1.5) & ; & 1 \leq x \leq 1.5 \\ 0.32 + 0.08(x - 2) & ; & 1.5 \leq x \leq 2 \\ 0.36 + 0.1(x - 2.5) & ; & 2 \leq x \leq 2.5 \end{cases}$$

At  $x = 2.25$  lie in  $2 \leq x \leq 2.5$

$$\begin{aligned} \therefore f(2.25) &= 0.36 + 0.1(2.25 - 2.5) \\ &= 0.36 - 0.025 \\ &= 0.335 \end{aligned}$$

Ans:-

Q. Given the following set of values of  $x$  and  $f(x)$ :  $(-1, 6)$ ,  $(1, 2)$  &  $(2, 12)$

A natural cubic is defined by

$$f(x) = \begin{cases} A(x+1)^3 + B(x+1)^2 + C(x+1) + 6 & ; & -1 \leq x \leq 1 \\ D(x-1)^3 + E(x-1)^2 + F(x-1) + 2 & ; & 1 \leq x \leq 2 \end{cases}$$

i Use Continuity and boundary Condition to estimate  $A, B, C, D, E$

ii Find the value of  $f(1.4)$  from cubic curve.

iii Write MATLAB codes using "sp = csape(x, y, 'second')" to Construct natural cubic spline for the above data set plot the spline curve using "fplot(sp)" along with the data.

(i)

Let,

$$f_1(x) = A(x+1)^3 + B(x+1)^2 + C(x+1) + 6 \quad \text{--- (1)}$$

$$f_2(x) = D(x-1)^3 + 6(x-1)^2 + E(x-1) + 2 \quad \text{--- (ii)}$$

Now, 1st and 2nd derivatives of (i) & (ii) =

$$f_1'(x) = 3A(x+1)^2 + 2B(x+1) + C$$

$$f_2'(x) = 3D(x-1)^2 + 12(x-1) + E$$

So  $f_1''(x) = 6A(x+1) + 2B$

$$f_2''(x) = 6D(x-1) + 12$$

At Common Condition  $x=1$  =

$$f_1(1) = f_2(1)$$

$$8A + 4B + 2C + 6 = 2$$

$$\Rightarrow 8A + 4B + 2C = 2 - 6$$

$$\Rightarrow 8A + 4B + 2C = -4$$

$$\Rightarrow 4A + 2B + C = -2 \quad \text{--- (iii)}$$

$$f_1'(1) = f_2'(1)$$

$$12A + 4B + C = E \quad \text{--- (iv)}$$

$$f_1''(1) = f_2''(1)$$

$$12A + 2B = 12$$

$$\Rightarrow 6A + B = 6 \quad \text{--- (v)}$$

For:- Natural cubic splines

We know:-

$$f_1''(x_0) = 0 \quad \& \quad f_2''(x_2) = 0$$

$$\therefore f_1''(-1) = 0 \quad \& \quad f_2''(2) = 0$$

$$\Rightarrow 6A(-1+1) + 2B = 0$$

$$\Rightarrow \boxed{B = 0}$$

$$\& \Rightarrow 6D(2-1) + 12 = 0$$

$$\Rightarrow D = -12/6 = -2$$

put the value of B in eq (v)  $\Rightarrow$

$$6A + B = 6$$

$$\Rightarrow 6A + 0 = 6$$

$$\therefore \boxed{A = 1}$$

put the value of A, B in eq (iii)  $\Rightarrow$

$$4A + 2B + C = -2$$

$$\Rightarrow C = -2 - 4A$$

$$\therefore C = -2 - 4 \times 1$$

$$\therefore \boxed{C = -6}$$

put the value of A, B, C in eq (iv)  $\Rightarrow$

$$12A + 4B + C = E$$

$$\therefore E = 12 \times 1 + 4 \times 0 + (-6)$$

$$\boxed{E = 6}$$

put all the value in given equation:

$$f(x) = \begin{cases} (x+1)^3 + 0 \cdot (x+1)^2 - 6(x+1) + 6, & -1 \leq x \leq 1 \\ -2(x-1)^3 + 6(x-1)^2 + 6(x-1) + 2, & 1 \leq x \leq 2 \end{cases}$$

ii

 $x = 1.4$  lies in  $1 \leq x \leq 2$ 

$$\therefore f_2(1.4) = -2(1.4-1)^3 + 6(1.4-1)^2 + 6(1.4-1) + 2$$

$$= 5.232$$

iii

⇒ MATLAB Codes:-  
 ~~~~~

» clear;

» $x = [-1, 1, 2];$

» $y = [6, 2, 12];$

» $sp = \text{csape}(x, y, 'spline');$

» $\text{solution} = \text{fnplt}(sp)$

problem 5:-
 ~~~~~

⇒ A natural cubic spline  $f(x)$  is defined by

$$f(x) = \begin{cases} A(x+1)^3 + B(x+1)^2 + C(x+1) - 6, & -1 \leq x \leq 1 \\ D(x-1)^3 + 12(x-1)^2 + E(x-1) + 2, & 1 \leq x \leq 3 \end{cases}$$

(i) Use continuity and boundary condition to find the value of  $A, B, C, D, E$

(ii) Estimate  $f(2)$  from the spline curve.

(iii) Use MATLAB function " $sp = \text{spline}(x, y)$ " and " $\text{fnval}(sp, x)$ " to construct the spline curve from the data  $(1, -10)$ ,  $(2, -6)$ ,  $(4, 2)$ ,  $(6, 15)$  find the values of  $f(x)$  for  $x = 2.4, 2.5, 4.8$  from spline curve.

①

Let  $f_1(x) = A(x+1)^3 + B(x+1)^2 + C(x+1) - 6$  ———— ①

$f_2(x) = D(x-1)^3 + 12(x-1)^2 + E(x-1) + 2$  ———— ②

Now int & 2nd derivatives of ① & ②  $\Rightarrow$

$f_1'(x) = 3A(x+1)^2 + 2B(x+1) + C$

$f_2'(x) = 3D(x-1)^2 + 24(x-1) + E$

Sol  $f_1''(x) = 6A(x+1) + 2B$

$f_2''(x) = 6D(x-1) + 24$

At Common point  $x=1$  :

| $f_1(x) = f_2(x) ; x=1$              | $f_1'(1) = f_2'(1)$            | $f_1''(1) = f_2''(1)$       |
|--------------------------------------|--------------------------------|-----------------------------|
| $\Rightarrow 8A + 4B + 2C - 6 = 2$   | $\Rightarrow 12A + 4B + C = E$ | $\Rightarrow 12A + 2B = 24$ |
| $\Rightarrow 8A + 4B + 2C = 8$       | ————— ④                        | $\Rightarrow 6A + B = 12$   |
| $\Rightarrow 4A + 2B + C = 4$ ———— ③ |                                | ————— ⑤                     |

For Natural Cubic Spline:-

We know:-

$$f_1''(x_0) = 0 \quad \& \quad f_2''(x_2) = 0$$

$$\therefore f_1''(-1) = 0 \quad \& \quad f_2''(3) = 0$$

$$\Rightarrow 6A(-1+1) + 2B = 0 \quad \& \Rightarrow 6D(3-1) + 24 = 0$$

$$\Rightarrow \boxed{B = 0}$$

$$\Rightarrow 12D = -24$$

$$\Rightarrow \boxed{D = -2}$$

put the value of B in eq (V)=

$$6A + B = 12$$

$$\therefore \boxed{A = 2}$$

put the value of A, B in eq (iii)=

$$4A + 2B + e = 4$$

$$\Rightarrow 8 + 0 + e = 4$$

$$\Rightarrow \boxed{e = -4}$$

put the value of A, B, C in eq (iv)=

$$E = 12A + 4B + e$$

$$\Rightarrow E = 12 \times 2 + 4 \times 0 - 4$$

$$\Rightarrow \boxed{E = 20}$$

Now put all the value in given equation:-

$$\therefore f(x) = \begin{cases} 2(x+1)^3 + 0(x+1)^2 - 4(x+1) - 6, & -1 \leq x \leq 1 \\ -2(x-1)^3 + 12(x-1)^2 + 20(x-1) + 2, & 1 \leq x \leq 3 \end{cases}$$

Ans:-

(ii)

$x=2$  lies at  $1 \leq x \leq 3$

$$\therefore f_2(x) = -2(x-1)^3 + 12(x-1)^2 + 20(x-1) + 2$$

$$\therefore f(2) = -2(2-1)^3 + 12(2-1)^2 + 20(2-1) + 2$$

$$= 32$$

(iii)

MATLAB CODE:-

```
>> clear;
>> x = [1, 2, 4, 6];
>> y = [-10, -6, 2, 15];
>> sp = spline(x, y)
>> x1 = [1.4, 2.5, 4.8];
>> y1 = fval(sp, x1)
```

Ans:-

# problems 6:—

⇒ A clamped cubic spline function  $f(x)$  through  $(1, 2)$ ,  $(2, 3)$ ,  $(3, 5)$  is defined by

$$f(x) = \begin{cases} 1.5(x-1)^3 + A(x-1)^2 + B(x-1) + 2, & 1 \leq x \leq 2 \\ C(x-2)^3 + 2(x-2)^2 + D(x-2) + 3, & 2 \leq x \leq 3 \end{cases}$$

Given that  $f'(1) = 2$  and  $f'(3) = 1$

- ① Use Continuity and boundary Conditions to find the values of  $A, B, C, D$
- ii Estimate  $f(2.5)$  from the spline curve.
- iii Use MATLAB function " $sp = \text{spline}(m, y)$ " and " $fval(sp, x)$ " to construct the spline curve from the given data and find the values of  $f(x)$  for  $x = 1.4$  and  $2.5$  from the spline curve.

①

Let,

$$f_1(x) = 1.5(x-1)^3 + A(x-1)^2 + B(x-1) + 2 \quad \text{--- ①}$$

$$f_2(x) = C(x-2)^3 + 2(x-2)^2 + D(x-2) + 3 \quad \text{--- ②}$$

Now,

1st and 2nd derivatives of eq ① & ② become:—

$$f'_1(x) = 4.5(x-1)^2 + 2A(x-1) + B$$

$$f'_2(x) = 3C(x-2)^2 + 4(x-2) + D$$

So  $f_1''(x) = 0(x-2) + 2A$

$f_2''(x) = 6C(x-2) + 4$

At Common value  $x = 2$  —

| $f_1(2) = f_2(2)$                                                                                            | $f_1'(2) = f_2'(2)$                   | $f_1''(2) = f_2''(2)$                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Rightarrow 1.5 + A + B + 2 = 3$<br>$\Rightarrow A + B = 3 - 2 - 1.5$<br>$\Rightarrow A + B = -0.5$ — (iii) | $\Rightarrow 4.5 + 2A + B = D$ — (iv) | $\Rightarrow 0 + 2A = 4$<br>$\Rightarrow A = \frac{4-0}{2}$<br><div style="border: 1px solid black; padding: 5px; display: inline-block;"> <math>A = -\frac{5}{2}</math> </div> |

For clamped spline cubic

We know:

$f_1'(x_0) = d_0$  &  $f_2'(x_2) = d_2$

$\therefore f_1'(2) = 2$  &  $f_2'(3) = 1$

$\Rightarrow 4.5(1-2)^2 + 2A(1-2) + B = 2$  &  $6C(3-2)^2 + 4(3-2) + D = 1$

$\therefore B = 2$

$\Rightarrow 6C + 4 + D = 1$  — (v)

put A and B in eq (iv) =

$\therefore D = 4.5 + 2A + B$

$= 4.5 + 2\left(-\frac{5}{2}\right) + 2$

$D = 1.5$

put D in eq (v)  $\Rightarrow$

$$3c + 4 + D = 1$$

$$\Rightarrow 3c = 1 - 4 - D$$

$$\Rightarrow c = \frac{1 - 4 - 1.5}{3}$$

$$\Rightarrow \boxed{c = -1.5}$$

Now put all the value in given eq:-

$$f(x) = \begin{cases} 1.5(x-1)^3 - \frac{5}{2}(x-1)^2 + 2(x-1) + 2, & 1 \leq x \leq 2 \\ -1.5(x-2)^3 + 2(x-2)^2 + 1.5(x-2) + 3, & 2 \leq x \leq 3 \end{cases}$$

Ans:-

(ii)

$x = 2.5$  lies in  $2 \leq x \leq 3$

$$\therefore f_2(x) = -1.5(x-2)^3 + 2(x-2)^2 + 1.5(x-2) + 3$$

$$\therefore f(2.5) = -1.5(2.5-2)^3 + 2(2.5-2)^2 + 1.5(2.5-2) + 3$$

$$= 4.0625$$

(iii)

MATLAB code:-

```
>> clear;
>> x=[2, 2.5];
>> y=[2, 3, 5];
>> sp=spline(x,y)
>> x1=[2.4, 2.5];
>> y1=fval(sp,x1)
```



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